

ω -Wide Diameters of (n, k) -Star Graphs, when $1 \leq k \leq \lfloor n/2 \rfloor$

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Abstract

Chiang and Chen proposed in 1995 the (n, k) -star graph which is a generalized version of the n -star graph. They also shown that the diameter of the (n, k) -star graph is $2k-1$ when $1 \leq k \leq \lfloor n/2 \rfloor$. To evaluate the fault-tolerant property of a graph, determining the wide diameter of the graph is a very importance issue. Lin and Duh (2008) computed upper and lower bounds of the wide diameter of the (n, k) -star graph, whose values are the same and its diameter plus 2 when $1 \leq k \leq \lfloor n/2 \rfloor$. This work further discuss ω -wide diameters, the general wide diameters, of the (n, k) -star graph for $2 \leq \omega \leq n-2$. By definition, the 1-wide diameter of the (n, k) -star graph is its diameter, and the $(n-1)$ -wide diameter is its wide diameter. Our result shows that ω -wide diameters of the (n, k) -star graph are its diameter for $1 \leq \omega \leq k-1$, diameter plus 1 for $k-1 < \omega \leq 2k-2$ and diameter plus 2 for $2k-2 < \omega \leq n-1$, respectively, when $1 \leq k \leq \lfloor n/2 \rfloor$ and diameter plus 2 for $2 \leq \omega \leq n-1$, when $k = 2$.

Keywords: (n, k) -star graphs; diameter; container; wide diameter; ω -wide diameters

1 Introduction

The topological structure of a massively parallel computer or an interconnection network is generally modeled by a graph. Moreover, graph theory is used to design and analyze topological properties of interconnection networks such as permutation capability [3], Hamiltonian laceability [2], fault tolerance [14], [16], [17], [34], performance [5], reliability [35] and combinatorial properties [18]. To evaluate reliability, fault-tolerant ability, maximum parallelism and minimum transmission delay of a graph (or an interconnection network), constructing vertex-disjoint paths and determining ω -wide diameters of the graph are very important issues [9], [13], [19], [20], [23], [33].

An (n, k) -star graph $(S_{n,k})$ was designed to be a generalized version of an n -star graph (S_n) , where

$S_{n,n-1}$ ($S_{n,1}$) and S_n (K_n) are isomorphic by the definition of an $S_{n,k}$ [11]. For building massively parallel computers, an S_n , which is a Cayley graph with a regular and hierarchical structure, has been recognized as an attractive alternative graph to an n -cube [1], [15]. An S_n has low degree, small diameter, scalability and symmetry [1]. Many topological properties of an S_n have been investigated such as parallel paths [9], [20], spanning trees [31] and a high degree of fault tolerance [21], [22], [26], [27], [32]. The number of vertices is $n!$ and additional $(n-1) \times (n-1)!$ vertices are required to expand an S_{n-1} to an S_n . Therefore, Chiang and Chen proposed the new graph $S_{n,k}$ to eliminate this restriction [11]. An $S_{n,k}$ also has many attractive properties such as vertex symmetry, hierarchical structure, maximal fault tolerance and simple shortest-path routing. Fortunately, only $(n-1)(n-1)!/(n-k)!$ additional vertices are required to expand an $S_{n-1,k-1}$ to an $S_{n,k}$. Some basic properties of an $S_{n,k}$, such as diameter [11], connectivity [11], broadcasting [10], average distance [12], embedding [6], [7], [8], Hamiltonicity [24], spanning connectivity [25], and wide diameter [28], have recently been computed or investigated. These measurement results indicate that an $S_{n,k}$ has excellent topological properties.

Let G denote a graph and $V(G)$ and $E(G)$ denote the vertex set and edge set of G , respectively. Two vertices u and v are *adjacent* when they are joined by an edge e , where $u, v \in V(G)$ and $e \in E(G)$. All vertices adjacent to a vertex are its *neighbors*. Let $P(u, v)$ ($P^*(u, v)$) denote a path (shortest path) from vertex u to vertex v , and $l(P(u, v))$ represent the length of $P(u, v)$. The *distance* from vertex u to vertex v , represented by $d(u, v)$, refers to $l(P^*(u, v))$. The *diameter* of G , denoted by $d(G)$, is defined as the maximum distance for all pairs of distinct vertices u and v in G [4]. A graph is *connected* when at least one path exists between any two vertices. The *connectivity* of a graph is defined as the minimum number of vertices whose removal renders it disconnected or trivial. Let κ be the connectivity of G . According to Menger's theorem, at least κ vertex-disjoint paths exist between distinct vertices u and v in G [4]. A set of ω vertex-disjoint paths between u and v , denoted by $C_\omega(u, v)$, is a *container* of width ω between u and v , where $1 \leq \omega \leq \kappa$. The

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length of a $C_\omega(u, v)$, denoted by $l(C_\omega(u, v))$, is defined as the length of the longest path in $C_\omega(u, v)$. A ω -best-container between u and v , denoted by $C_\omega^*(u, v)$, is the container among all $C_\omega(u, v)$ s, with the shortest length. Let $d_\omega(u, v)$ indicate the ω -wide distance (or wide distance) from u to v ; thus, $d_\omega(u, v) = l(C_\omega^*(u, v))$ [23]. The ω -wide diameter of G , denoted by $d_\omega(G)$, is defined as the maximum of $d_\omega(u, v)$ s for all pairs of distinct vertices u and v in G .

Chiang and Chen (1995) showed that the connectivity of an $S_{n,k}$ is $n-1$ [11]. Hence, by Merger's theorem, there are at least $n-1$ vertex-disjoint paths between any distinct vertices u and v in an $S_{n,k}$. Lin, Duh and Cheng presented in 2004 that $d_{n-1}(S_{n,n-1}) = d(S_{n,n-1}) + 2$ and $d_{n-1}(S_{n,1}) = d(S_{n,1}) + 1$. Recently, Lin and Duh (2008) demonstrated that $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 2$ if $2 \leq k \leq \lfloor n/2 \rfloor$, $d_4(S_{5,3}) = d(S_{5,3}) + 2$, $d_5(S_{6,4}) = d(S_{6,4}) + 2$, and $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 1$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \notin \{5, 6\}$ [28], [30]. Lin and Duh provided algorithms to construct $C_{n-1}(u, v)$ where u and v are any two distinct vertices in an $S_{n,k}$ [28]. The lengths of some of the $C_{n-1}(u, v)$ s constructed by Lin and Duh are not greater than $d(S_{n,k})$. This work focuses on those constructed $C_{n-1}(u, v)$ s whose lengths are greater than $d(S_{n,k})$. By building a $C_\omega^*(u, v)$ between each selected pair of vertices in an $S_{n,k}$, $d_\omega(S_{n,k})$ can thus be determined. This work shows that $d_\omega(S_{n,2}) = d(S_{n,2}) + 2$ for $2 \leq \omega \leq n-1$, $d_\omega(S_{n,k}) = d(S_{n,k})$ for $2 \leq \omega \leq k-1$ and $3 \leq k \leq \lfloor n/2 \rfloor$, $d_\omega(S_{n,k}) = d(S_{n,k}) + 1$ for $k-1 < \omega \leq 2k-2$ and $3 \leq k \leq \lfloor n/2 \rfloor$, $d_\omega(S_{n,k}) = d(S_{n,k}) + 2$ for $2k-2 < \omega \leq n-1$ and $3 \leq k \leq \lfloor n/2 \rfloor$.

The remainder of this paper is organized as follows. Section 2 formally describes the topology and properties of an $S_{n,k}$ and some notations frequently used in this work. Section 3 constructs paths between selected pairs of vertices in an $S_{n,k}$ and shows the constructed paths are all vertex-disjoint. Section 4 computes $d_\omega(S_{n,k})$ by calculating the length of each constructed path. Conclusions are finally drawn in Section 5.

2 Background & Notations

According to the structure of an $S_{n,k}$, it is an undirected-regular and vertex-symmetrical graph, where parameters n and k are utilized to determine the topological properties, such as number of vertices, connectivity and diameter of an $S_{n,k}$ [11]. A vertex ρ can be denoted as a label $\rho_1\rho_2\ldots\rho_k$ and the vertex set $V(S_{n,k})$ is defined as $\{\rho_1\rho_2\ldots\rho_k \mid \rho_i \neq \rho_j, \rho_i \text{ and } \rho_j \text{ in } \langle n \rangle\}$. Each element in $\langle n \rangle$ is a symbol and every symbol $\in \langle k \rangle$ ($\in \langle n \rangle - \langle k \rangle$) is an *internal* (*external*) symbol. Moreover, symbol ρ_i in label $\rho_1\rho_2\ldots\rho_k$ can be: (1) *fixed* if $\rho_i = i$ (notably, position i is the *desired position* for symbol i); (2) *misplaced* if $\rho_i \neq i$ and ρ_i is

an internal symbol. A symbol not in label $\rho_1\rho_2\ldots\rho_k$ is *desired* if it is an internal symbol and *outer external* if it is an external symbol. For example, a vertex $\rho = 489236$ in $S_{9,6}$, $\rho_6 = 6$ is a fixed symbol. Additionally, $\rho_1 = 4$, $\rho_4 = 2$ and $\rho_5 = 3$ are misplaced symbols; symbols 1 and 5 are desired symbols; symbols 8 and 9 are external symbols, and symbol 7 is an outer external symbol.

Vertex ρ is adjacent to three kinds of neighbors ρ^i , ρ^1 and $\rho^{1'}$ which are defined as follows:

1. $\rho^i = \rho_1\rho_2\ldots\rho_{i-1}\rho_{i+1}\ldots\rho_k$, for all $2 \leq i \leq k$ (swapping ρ_1 with ρ_i).
2. $\rho^1 = x\rho_2\ldots\rho_k$, for all x is a desired symbol (replacing ρ_1 with x).
3. $\rho^{1'} = y\rho_2\ldots\rho_k$, for all y is an outer external symbol (replacing ρ_1 with y).

In other words, ρ^i , ρ^1 , and $\rho^{1'}$ represent i -, 1 -, and $1'$ -neighbors of vertex ρ , respectively. Based on the above definition, $\rho^{\alpha\beta}$ is used to denote β -neighbors of each vertex ρ^α ; $\rho^{\alpha\beta\gamma}$ is used to denote γ -neighbors of each vertex $\rho^{\alpha\beta}$, where α, β and $\gamma \in \{i, 1, 1'\}$, and so on. Additionally, let ρ_j^α represent the j th symbol of vertex $\rho^\alpha = \rho_1^\alpha\rho_2^\alpha\ldots\rho_k^\alpha$, where $1 \leq j \leq k$ and $\alpha \in \{i, 1, 1'\}$.

Fig. 1 shows the structure of $S_{4,2}$, and Fig. 2 illustrates all neighbors of vertex 8263 in $S_{8,4}$.

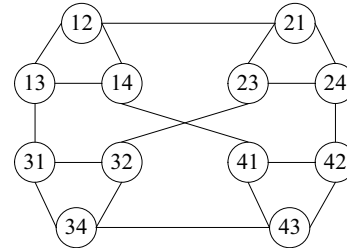


Fig. 1. The structure of $S_{4,2}$.

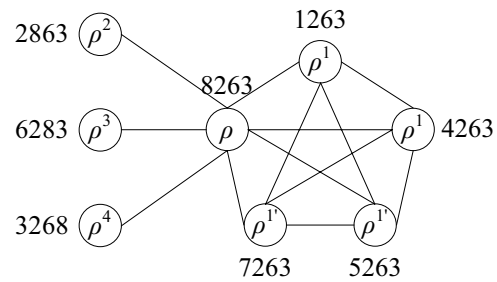


Fig. 2. All neighbors of vertex 8263 in $S_{8,4}$.

Vertex $\varepsilon = 12\ldots k$ is an *identity* vertex whose $\varepsilon_i = i$ for all $1 \leq i \leq k$. The vertex symmetry of $S_{n,k}$ [11] allows us to assume that ε is the destination vertex. Lin and Duh provided algorithms to construct $n-1$ vertex-disjoint paths from vertex ρ to ε to form a $C_{n-1}(\rho, \varepsilon)$ [28]. This work attempts to find a $C_\omega^*(\rho, \varepsilon)$ for $2 \leq \omega \leq n-1$. Notably, it is not necessary to find out a $C_\omega^*(\rho, \varepsilon)$ for every vertex ρ in $S_{n,k}$, because the lengths of some $C_{n-1}(\rho, \varepsilon)$ s constructed by Lin and Duh are smaller than $d(S_{n,k})$ which is the trivial lower bound of $d_\omega(S_{n,k})$. In other words, some vertices in

$S_{n,k}$ are not critical for determining $d_\omega(S_{n,k})$. Therefore, this work only considers every vertex ρ whose $l(C_{n-1}(\rho, \varepsilon))$ is greater than $d(S_{n,k})$, where $C_{n-1}(\rho, \varepsilon)$ is built by Lin and Duh. Hence, based on Lin and Duh's algorithm, all vertices in $S_{n,k}$ are divided into three groups: A vertex ρ is in Group 1 (Group2) if $l(C_\omega(\rho, \varepsilon)) > d(S_{n,k})$ and $p_1=1$ ($p_1 \neq 1$)), and it refers to Case 1 (Case 2) as shown in Table 1. A vertex ρ is in Group 3 if $l(C_\omega(\rho, \varepsilon)) \leq d(S_{n,k})$ and will be ignored in calculating $d_\omega(S_{n,k})$. As mentioned above, Cases 1 and 2 are further separated to more sub-cases according to the numbers of external symbols as described in Table 1.

Table 1. Considered Cases

Cases		Number of external symbols	Number of misplaced symbols	Length of container built by Lin
Case 1 ($p_1 = 1$)	Case 1.1	$k-1$	0	$2k+1$
	Case 1.2	$k-2$	1	$2k$
	Case 1.3	$k-3$	2	$2k$
Case 2 ($p_1 \neq 1$)	Case 2.1	k	0	$2k+3$
	Case 2.2	$k-1$	0 or 1	$2k+2$
	Case 2.3	$k-2$	1	$2k$

For constructing ω vertex-disjoint paths from vertex ρ to ε , all of the paths have to pass through neighbors of ρ and neighbors of ε once at last. Consequently, neighbors of ρ to neighbors of ε are one-to-one, and every constructed path has to vertex-disjoint to the others. Generally, the starting vertex is ρ and the destination vertex is ε . Significantly, the most important issue is to determine a vertex pair of neighbors of ρ and ε . Symbol $\rightarrow$ is used to represent the edge from a vertex to its successor vertex in a path. Moreover, to represent a path from a vertex ρ by fixing symbols $\rho_x, \rho_y, \rho_z, \dots$ to be $x, y, z, \dots$ orderly, which are all internal, use a correcting sequence, denoted by $\langle x, y, z, \dots \rangle$, to indicate the sequence. For example, if $\rho = 6789$ in $S_{9,4}$, and the correcting sequence is $\langle 2, 3, 4 \rangle$, then the path is constructed as: $6789 \rightarrow 2789 \rightarrow 7289 \rightarrow 3289 \rightarrow 8239 \rightarrow 4239 \rightarrow 9234$. A path from vertex u to vertex v constructed according to correcting sequence $\langle x, y, z, \dots \rangle$ is briefly denoted as $u \rightarrow \langle x, y, z, \dots \rangle v$. For example, $6789 \rightarrow 2789 \rightarrow 7289 \rightarrow 3289 \rightarrow 8239 \rightarrow 4239 \rightarrow 9234$ can also denoted as $6789 \rightarrow \langle 2, 3, 4 \rangle 9234$. Notably, all symbols in a correcting sequence are all distinct. In other words, $\langle 2, \dots, k \rangle$, which excludes symbols j, g and i , is a subsequence of $\langle j, g, 2, \dots, k, i \rangle$. For simplicity, let $P(u, v) \rightarrow P(x, y)$ represent the concatenation of $P(u, v)$ and $P(x, y)$, where $v \neq x$. Also, let $P(u, w, v)$ ($P(u, s, t, v)$) be a path from vertex u passing through vertex w (vertices s and t) to vertex v .

3 Construction of Vertex-Disjoint Paths

This section first constructs $n-1$ paths from ρ to ε in an $S_{n,k}$, and then shows the constructed paths are vertex-disjoint case by case. Actually, the constructed paths are composed of 3 groups of paths: The first group paths passes through ρ^1 ; the second group of paths passes through ρ^i , and the third group of paths passes through $\rho^{1'}$. This section first proves that paths in a group are vertex-disjoint individually and then shows that these 3 groups of paths are all vertex-disjoint to each other. To prove each group of paths with some property and cross groups are vertex-disjoint, Lemmas 1, 2 and 3 are specifically provided for different properties.

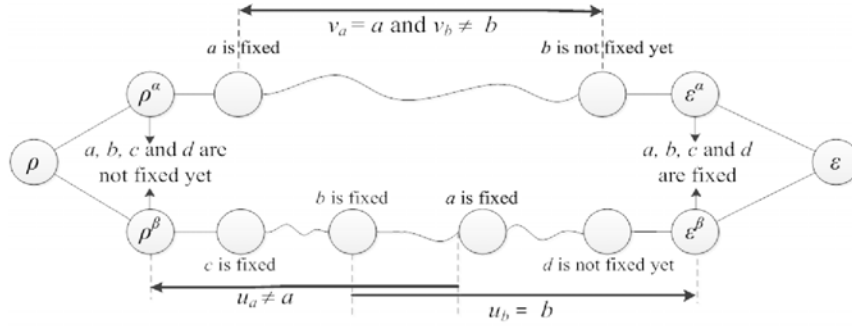
Lemma 1. Given two correcting sequences, the first (last) symbol of correcting sequences are a and c (b and d), respectively, and the order of b in the second correcting sequence is prior to a . If $P(\rho^a, \varepsilon^a) = \rho^a \rightarrow \langle a, \dots, c, \dots, d, \dots, b \rangle \varepsilon^a$ and $P(\rho^b, \varepsilon^b) = \rho^b \rightarrow \langle c, \dots, b, \dots, a, \dots, d \rangle \varepsilon^b$ are two distinct paths, where a and $b \in \{i, 1, 1'\}$, then $P(\rho^a, \varepsilon^a)$ and $P(\rho^b, \varepsilon^b)$ are vertex-disjoint to each other.

Proof. As shown in Fig. 3, $\rho_a^a \neq a$ and $\rho_b^b \neq b$ ($\varepsilon_a^a = a$ and $\varepsilon_b^b = b$) and each vertex $v \in V(P(\rho^a, \varepsilon^a)) \setminus \{\rho^a, \varepsilon^a\}$ has $v_a = a$ and $v_b \neq b$ and each vertex $w \in V(P(\rho^b, \varepsilon^b)) \setminus \{\rho^b, \varepsilon^b\}$ has $w_c = c$ and $w_d \neq d$. Naturally, ρ^a and ε^a are vertex-disjoint to each vertex v . Since $\rho_c^a \neq c$ ($\varepsilon_d^a = d$), ρ^a and ε^a are also vertex-disjoint to $P(\rho^b, \varepsilon^b)$. Furthermore, each vertex v is vertex-disjoint to each vertex $u \in V(P(\rho^b, \varepsilon^b))$ because u does not have $u_a = a$ and $u_b \neq b$. Therefore, $P(\rho^a, \varepsilon^a)$ and $P(\rho^b, \varepsilon^b)$ are vertex-disjoint to each other. $\square$

Lemma 2. Given two distinct paths $P(p, q) = P(p, p') \rightarrow P(q', q)$ and $P(u, v) = P(u, u') \rightarrow P(v', v)$, let every vertex in $P(p, p')$, $P(q', q)$, $P(u, u')$ and $P(v', v)$ denote by s, t, w and x , respectively. If $s_a = w_b = \sigma$, $s_b \neq 1$, $w_a \neq 1$ and $t_a = x_b = 1$, where $2 \leq \sigma \leq n$, $2 \leq a, b \leq k$ and $a \neq b$, then $P(p, q)$ and $P(u, v)$ are vertex-disjoint to each other.

Proof. Since a vertex having σ at both positions a and b is impossible and $s_a = w_b = \sigma$, every vertex s in $P(p, p')$ is distinct from every vertex w in $P(u, u')$. Similarly, every vertex t in $P(q', q)$ is distinct from every vertex x in $P(v', v)$. Moreover, every vertex s in $P(p, p')$ (w in $P(u, u')$) is distinct from every vertex x in $P(v', v)$ (t in $P(q', q)$) because $s_b \neq 1$ and $x_b = 1$ ($w_a \neq 1$ and $t_a = 1$). Therefore, $P(p, q)$ and $P(u, v)$ are vertex-disjoint to each other. $\square$

Lemma 3. Assuming P_1 and P_2 are two paths. If every vertex in P_1 contains symbol σ and every vertex in $P(u, v)$ does not contain symbol σ , then P_1 and P_2 are vertex-disjoint to each other.

Fig. 3. $P(\rho^\alpha, \varepsilon^\alpha)$ and $P(\rho^\beta, \varepsilon^\beta)$ are vertex-disjoint.

Proof. Since every vertex w in P_1 contains symbol σ and every vertex x in P_2 does not contain symbol σ , vertices w and x are distinct. Therefore, P_1 and P_2 are vertex-disjoint to each other. $\square$

A method for constructing vertex-disjoint paths is to circular shift a correcting sequence. For example, given a correcting sequence $\langle a, b, c \rangle$, two different correcting sequences $\langle b, c, a \rangle$, $\langle c, a, b \rangle$ can be obtained by circular left shifting the original correcting sequence 1 and 2 symbol(s). In the rest of this paper, for simplicity, let $i \ominus a$ denote $i - a$ ($k - a + i - 1$) if $i \geq a + 2$ ($i < a + 2$) and $i \oplus a$ denote $i + a$ ($i + a + 1 - k$) if $i \leq k - a$ ($i > k - a$), where $1 \leq a \leq k - 2$.

Since Lin and Duh have provided algorithms to build $C_{n-1}(\rho, \varepsilon)$ for each vertex ρ [28], this work only takes care of each ρ with $l(C_{n-1}(\rho, \varepsilon)) > d(S_{n,k})$. Conceptually, to build a $C_{\omega}^*(\rho, \varepsilon)$ for each selected ρ , first construct $P(\rho, \varepsilon)$ s with a length not greater than $d(S_{n,k})$ as many as possible; second construct $P(\rho, \varepsilon)$ s with a length of $d(S_{n,k}) + 1$ as many as possible; third construct $P(\rho, \varepsilon)$ s with a length of $d(S_{n,k}) + 2$ as many as possible, and so on. Notably, a $P(\rho, \varepsilon)$ with a length greater than $d(S_{n,k})$ may not exist and the number of shorter $P(\rho, \varepsilon)$ s may affect the number of longer $P(\rho, \varepsilon)$ s. In the rest of this section, a path construction is provided for each case as shown in Table 1. After building paths, this section also shows that the constructed paths are all vertex-disjoint. In the following, Sections 3.1 to 3.6 are used to describe Cases 1.1, 1.2, 1.3, 2.1, 2.2, and 2.3, respectively. Notably, in the rest of this work, the description about vertex-disjoint excludes vertices ρ and ε .

3.1 Case 1.1: $\rho_1 = 1$ and every ρ_i is external for $2 \leq i \leq k$

Let ρ be $1e_2e_3\dots e_k$ and the original correcting sequence be $\langle 2, 3, \dots, k \rangle$. There are $k-1$ different correcting sequences $\langle i, i \oplus 1, \dots, k, 2, 3, \dots, i \ominus 1 \rangle$ by circular shifting $\langle 2, 3, \dots, k \rangle$ $i-2$ times and each of them is assigned to ρ^1 with $\rho_1^1 = i$. Hence, $k-1$ $P(\rho^1, \varepsilon^1)$ s with $\varepsilon_1^1 = \rho_{i \oplus 1}^1$ can be constructed as $\rho^1 \rightarrow \langle i, i \oplus 1, \dots, k, 2, 3, \dots, i \ominus 1 \rangle \varepsilon^1$. According to Lemma 1, the constructed $k-1$ $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint. Moreover, each of $k-1$ $P(\rho^1, \varepsilon)$ s can be constructed as

$\rho^1 \rightarrow \langle i \oplus 1, i \oplus 2, \dots, k, 2, \dots, i \ominus 1, i \ominus 1 \ominus 1 \rangle \varepsilon^1$. Due to symbol 1 is at position i in each vertex of each $P(\rho^1, \varepsilon^1)$, the constructed $k-1$ $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint. Notably, there are $n-2k+1$ different outer external symbols, each of them denoted by e' . Thus, $n-2k+1$ $P(\rho^{1'}, \rho^{1'j}, \varepsilon^{1'})$ s with $\rho_1^{1'} = e'$ and $\varepsilon_1^{1'} = \rho_{i \oplus 1}^{1'}$ can be built as $\rho^{1'} \rightarrow \rho^{1'j} \rightarrow \langle 2, 3, \dots, i \ominus 1, i \oplus 1, \dots, k, i \oplus 1 \rangle \varepsilon^{1'}$, where $2 \leq j \leq k$. With the aid of Lemma 3, the constructed $n-2k+1$ $P(\rho^{1'}, \rho^{1'j}, \varepsilon^{1'})$ s are vertex-disjoint.

Each $P(\rho^i, \varepsilon^i)$ is vertex-disjoint to each $P(\rho^1, \varepsilon^1)$, where $\varepsilon_1^1 = \rho_{i \oplus 1}^1$, because every vertex in $P(\rho^i, \varepsilon^i)$ has symbol 1 on its position i but every vertex in $P(\rho^1, \varepsilon^1)$ has not. According to Lemma 3, each $P(\rho^{1'}, \varepsilon^{1'})$, where $\varepsilon_1^{1'} = \rho_1^{1'}$, is vertex-disjoint to each $P(\rho^i, \varepsilon^i)$ and each $P(\rho^1, \varepsilon^1)$, where $\varepsilon_1^1 = \rho_{i \oplus 1}^1$. Thus, $k-1$ $P(\rho^1, \varepsilon^1)$ s, $k-1$ $P(\rho^i, \varepsilon^i)$ s, and $n-2k+1$ $P(\rho^{1'}, \varepsilon^{1'})$ s are all vertex-disjoint. Therefore, after adding edges $\rho \rightarrow \rho^1$, $\rho \rightarrow \rho^i$, $\rho \rightarrow \rho^{1'}$, $\varepsilon^1 \rightarrow \varepsilon$, and $\varepsilon^i \rightarrow \varepsilon$ to their corresponding $P(\rho^1, \varepsilon^1)$ s, $P(\rho^i, \varepsilon^i)$ s, and $P(\rho^{1'}, \varepsilon^{1'})$ s, a $C_{n-1}(\rho, \varepsilon)$ can be formed.

3.2 Case 1.2: $\rho_1 = 1$, $\exists \rho_j$ is misplaced and every ρ_i is external for $2 \leq i, j \leq k$ and $i \neq j$

Since the numbers of ρ^1 and ρ^j are $k-2$ and $k-1$, respectively, $k-2$ $P(\rho^1, \varepsilon^1)$ s with $\varepsilon_1^1 = \rho_{i \oplus 1}^1$ and $k-1$ $P(\rho^j, \varepsilon^j)$ s can be constructed similar to Case 1.1. Let e' denote an outer external symbol. Every $P(\rho^{1'}, \rho^{1'j}, \varepsilon^{1'})$ s with $\rho_1^{1'} = e'$ and $\varepsilon_1^{1'} = \rho_{i \oplus 1}^{1'}$ can be built as $\rho^{1'} \rightarrow \rho^{1'j} \rightarrow \langle \rho_j, j \oplus 1, \dots, k, 2, 3, \dots, i \ominus 1 \rangle \varepsilon^{1'}$. In total, $n-2k+2$ $P(\rho^{1'}, \rho^{1'j}, \varepsilon^{1'})$ s can be built. With the aid of Lemma 3, the constructed $n-2k+2$ $P(\rho^{1'}, \rho^{1'j}, \varepsilon^{1'})$ s are vertex-disjoint.

That each $P(\rho^i, \varepsilon^i)$ is vertex-disjoint to each $P(\rho^1, \varepsilon^1)$ can be similarly proved as that in Case 1.1. According to Lemma 3, each $P(\rho^{1'}, \rho^{1'j}, \varepsilon^{1'})$, where $\varepsilon_1^{1'} = \rho_1^{1'}$, is vertex-disjoint to each $P(\rho^i, \varepsilon^i)$ and each $P(\rho^1, \varepsilon^1)$, where $\varepsilon_1^1 = \rho_{i \oplus 1}^1$. Therefore, $k-2$ $P(\rho^1, \varepsilon^1)$ s, $k-1$ $P(\rho^j, \varepsilon^j)$ s, and $n-2k+2$ $P(\rho^{1'}, \varepsilon^{1'})$ s are all vertex-disjoint and a $C_{n-1}(\rho, \varepsilon)$ can thus be formed.

3.3 Case 1.3: $\rho_1 = 1$, $\exists \rho_j$ and ρ_g , $j \neq g$, are misplaced and every ρ_i is external for $2 \leq i, j, g \leq k$ and $i \notin \{j, g\}$

Since the numbers of ρ^1 and ρ^j are $k-3$ and $k-1$,

respectively, $k-3$ $P(\rho^1, \varepsilon^1)$ s with $\varepsilon_1' = \rho_{i \oplus 1}$ and $k-1$ $P(\rho^i, \varepsilon^i)$ s can be similarly constructed as that in Case 1.1. Let e' denote an outer external symbol. Every $P(\rho^1, \rho^{1j}, \varepsilon^1)$ with $\rho_1' = e'$ and $\varepsilon_1' = \rho_1'$ can be built in three cases which depend on ρ_j and ρ_g . Case a. ($\rho_j = g$ and $\rho_g = j$): $\rho^1 \rightarrow \rho^{1j} \rightarrow_{\langle g, j \rangle} \rho^{1j, g, j} \rightarrow_{\langle i \oplus 1, \dots, k, 2, 3, \dots, j \oplus 1 \rangle} \varepsilon^1$. Case b. ($\rho_j = g$ and $\rho_g = i$): $\rho^1 \rightarrow \rho^{1j} \rightarrow_{\langle g, i, j \oplus 1, \dots, k, 2, 3, \dots, j \oplus 1 \rangle} \varepsilon^1$. Case c. ($\rho_j = i$ and $\rho_g = i'$, where $2 \leq i' \leq k$ and $i' \notin \{i, j, g\}$): $\rho^1 \rightarrow \rho^{1j} \rightarrow_{\langle i, g, i', j \oplus 1, \dots, k, 2, 3, \dots, j \oplus 1 \rangle} \varepsilon^1$. In total, $n-2k+3$ $P(\rho^1, \rho^{1j}, \varepsilon^1)$ s can be built. With the aid of Lemma 3, the constructed $n-2k+3$ $P(\rho^1, \rho^{1j}, \varepsilon^1)$ s are vertex-disjoint.

That each $P(\rho^i, \varepsilon^i)$ is vertex-disjoint to each $P(\rho^1, \varepsilon^1)$ can be similarly proved as that in Case 1.1. According to Lemma 3, each $P(\rho^1, \rho^{1j}, \varepsilon^1)$ with $\varepsilon_1' = \rho_1'$ is vertex-disjoint to each $P(\rho^i, \varepsilon^i)$ and each $P(\rho^1, \varepsilon^1)$ with $\varepsilon_1' = \rho_{i \oplus 1}$. Therefore, $k-3$ $P(\rho^1, \varepsilon^1)$ s, $k-1$ $P(\rho^i, \varepsilon^i)$ s, and $n-2k+3$ $P(\rho^1, \varepsilon^1)$ s are all vertex-disjoint and a $C_{n-1}(\rho, \varepsilon)$ can thus be formed.

3.4 Case 2.1: ρ_1 and every ρ_i are external for $2 \leq i \leq k$

The construction of each path is described below depending on the value of ω . For $1 \leq \omega \leq k-1$, since ρ has k different ρ^1 s with $\rho_1' \in \langle k \rangle$, $k-1$ vertex-disjoint $P(\rho^1, \varepsilon^1)$ s with $\rho_1' \neq 1$ and $\varepsilon_1' = \rho_{i \oplus 1}$ can be similarly constructed and proved as that in Case 1.1. Therefore, a $C_{k-1}(\rho, \varepsilon)$ can be formed.

For $k \leq \omega \leq n-1$, $n-2k$ vertex-disjoint $P(\rho^1, \rho^{1j}, \varepsilon^1)$ s with $\rho_1' = e'$ and $\varepsilon_1' = \rho_1'$ can be similarly constructed and proved as that in Case 1.1. However, the constructions of $P(\rho^1, \varepsilon)$ s and $P(\rho^i, \varepsilon)$ s should be reconsidered because they are not the same as that in Case 1.1. For showing the paths in different constructions are vertex-disjoint clearly, constructed paths are therefore divided into four groups as follows.

- i Totally, $k-1$ $P(\rho, \rho^1, \varepsilon)$ s with lengths no more than $d(S_{n,k})+1$ are built as follows. For simplicity, let $x = \rho_1^1$. $k-4$ $P(\rho, \rho^1, \varepsilon)$ s for $3 \leq x \leq k-2$ are constructed as $\rho^1 \rightarrow \rho^{1,x} \rightarrow \rho^{1,x,1} \rightarrow \rho^{1,x,1,x \oplus 1} \rightarrow_{\langle x \oplus 2, \dots, k, 2, \dots, x \oplus 1, x \oplus 1 \rangle} \varepsilon$ with $\rho_1^{1,x,1} = 1$. For $x = k \oplus 1$, a $P(\rho, \rho^1, \varepsilon)$ is constructed as $\rho^1 \rightarrow \rho^{1,k \oplus 1} \rightarrow \rho^{1,k \oplus 1,1} \rightarrow_{\langle 2, \dots, k \oplus 2, k \oplus 2 \rangle} \varepsilon$ with $\rho_1^{1,k \oplus 1,1} = 1$. For $x = k$, a $P(\rho, \rho^1, \varepsilon)$ is constructed as $\rho^1 \rightarrow \rho^{1,k} \rightarrow \rho^{1,k,1} \rightarrow \rho^{1,k,1,2} \rightarrow_{\langle 3, \dots, k \oplus 1, 2 \rangle} \varepsilon$ with $\rho_1^{1,k,1} = 1$. For $x = 1$, a $P(\rho, \rho^1, \varepsilon)$ is constructed as $\rho^1 \rightarrow \rho^{1,3} \rightarrow_{\langle 4, \dots, k, 2, 3 \rangle} \varepsilon$. Due to each path has unique ρ^1 , $\rho^{1,x}$ and $\rho^{1,x,1}$, where $1 \leq x \leq k$ and $x \neq 2$, and symbol 1 is in vertices at different position for each path after $\rho^{1,x,1}$, the $k-1$ constructed $P(\rho, \rho^1, \varepsilon)$ s are vertex-disjoint.
- ii Path $P(\rho^1, \varepsilon)$ with $x=2$ and $\varepsilon_1' = \rho_k$ is constructed as $\rho^1 \rightarrow_{\langle 2, 3, \dots, k \rangle} \varepsilon$.
- iii Path $P(\rho^2, \varepsilon)$ with $\varepsilon_1' = \rho_2$ is constructed as $\rho^2 \rightarrow_{\langle 3, 4, \dots, k, 2 \rangle} \varepsilon$.

- iv Totally, $k-2$ $P(\rho^h, \varepsilon)$ s with $\varepsilon_1' = \rho_{h-1}$, where $3 \leq h \leq k$, are constructed as $\rho^h \rightarrow_{\langle h, h \oplus 1, \dots, k, 2, \dots, h \oplus 1 \rangle} \varepsilon$. According to Lemma 1, $k-2$ $P(\rho^h, \varepsilon^1)$ s are vertex-disjoint.

Each path of **i** is vertex-disjoint to each path in **ii**, **iii** and **iv**, because every path in **ii**, **iii** and **iv** does not contain ρ^1 and $\rho^{1,x}$ and each vertex in those paths does not contain symbol 1. The path of **ii** is vertex-disjoint to the path of **iii**, because the symbol ρ_1 is in vertices of **iii** but not in any vertex of **ii**. The path of **ii** is vertex-disjoint to each path in **iv**, according to Lemma 1 and no ρ^h in **ii**. The path of **iii** is vertex-disjoint to each path in **iv**, because ρ^h is unique and symbol ρ_1 is in every vertex of **iii** but not in any vertex of **iv**. Thus, all paths in **i**, **ii**, **iii** and **iv** are vertex-disjoint. Moreover, each path of $P(\rho^1, \varepsilon^1)$ s is also vertex-disjoint to each path in **i**, **ii**, **iii** and **iv** by Lemma 3. Therefore a $C_{n-1}(\rho, \varepsilon)$ can thus be formed.

3.5 Case 2.2: $\rho_1 \neq 1$, and ρ has $k-1$ external symbols

The indefinite symbol ρ_1 is misplaced or ρ_j can be a fixed symbol, symbol 1 or misplaced symbol. Thus, three cases should be considered as follows.

3.5.1 $\exists \rho_j$ is fixed, and ρ_1 and every ρ_i are external for $2 \leq i, j \leq k$ and $i \neq j$

Let the original correcting sequence be $\langle 2, 3, \dots, j \oplus 1, j \oplus 1, \dots, k \rangle$. There are $k-2$ different correcting sequences $\langle i, i \oplus 1, \dots, k, 2, 3, \dots, j \oplus 1, j \oplus 1, \dots, i \oplus 1 \rangle$ by circular shifting $\langle 2, 3, \dots, j \oplus 1, j \oplus 1, \dots, k \rangle$ $i-2$ ($i-1$) times if $i < j$ ($i > j$) and each of them is assigned to ρ^1 with $\rho_1' = i$. Hence, $k-2$ $P(\rho^1, \varepsilon^1)$ s with $\varepsilon_1' = \rho_{i \oplus 1}$ can be constructed as $\rho^1 \rightarrow_{\langle i, i \oplus 1, \dots, k, 2, 3, \dots, j \oplus 1, j \oplus 1, \dots, i \oplus 1 \rangle} \varepsilon^1$. According to Lemma 1, the constructed $k-2$ $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint. One $P(\rho^1, \varepsilon^1)$ with $\rho_1' = 1$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow_{\langle j \oplus 1, \dots, k, 2, \dots, j \oplus 1, j \oplus 1 \rangle} \varepsilon$, and it is vertex-disjoint to each $P(\rho^1, \varepsilon^1)$ because symbol 1 is in every vertex of it but not in any vertex of each $P(\rho^1, \varepsilon^1)$.

One $P(\rho^j, \varepsilon^1)$ with $\varepsilon_1' = \rho_1$ is constructed as $\rho \rightarrow \rho^j \rightarrow_{\langle j \oplus 1, \dots, k, 2, \dots, j \oplus 1 \rangle} \varepsilon^1$. Since each ρ^j has a fixed ρ_j , j should not be in the correcting sequence of it. $k-2$ $P(\rho^i, \varepsilon)$ s are constructed as $\rho \rightarrow \rho^i \rightarrow_{\langle i \oplus 1, \dots, k, 2, \dots, i \oplus 1 \rangle} \varepsilon$ with $\rho_1^{s,1} = 1$, where $s = i, 1, i \oplus 1, \dots, 1, k, 1, 2, \dots, 1, i \oplus 1$. Since every vertex u in each $P(\rho^i, \varepsilon)$ has $u_i = \rho_1$ or $u_i = 1$, by Lemma 2, $k-2$ $P(\rho^i, \varepsilon)$ s are vertex-disjoint. Moreover, $k-2$ $P(\rho^i, \varepsilon)$ s are vertex-disjoint to $k-1$ $P(\rho^1, \varepsilon)$ s, because every vertex of each $P(\rho^1, \varepsilon)$ does not has ρ_1 , but every vertex of each $P(\rho^i, \varepsilon)$ has ρ_1 . The $P(\rho^j, \varepsilon^1)$ is vertex-disjoint to each other paths built before, because every vertex of it has ρ_1 and ρ_j which is not fixed, but no vertex of other paths has that. Therefore, $P(\rho^i, \varepsilon)$ s, $P(\rho^j, \varepsilon^1)$ and $P(\rho^1, \varepsilon)$ s are vertex-disjoint.

In total, $n-2k+1$ $P(\rho^{1'}, \rho^{1',i}, \varepsilon^{1'})$ s with $\varepsilon_1^{1'} = \rho_1^{1'}$ are build as $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1',i} \rightarrow \dots \rightarrow \rho^{1',j+1} \rightarrow \dots \rightarrow \rho^{1',k} \rightarrow \varepsilon^{1'}$. With the aid of Lemma 3, the constructed $P(\rho^{1'}, \rho^{1',i}, \varepsilon^{1'})$ s are vertex-disjoint, and each $P(\rho^{1'}, \rho^{1',i}, \varepsilon^{1'})$ is vertex-disjoint to $P(\rho^i, \varepsilon)$ s, $P(\rho^j, \varepsilon^{1'})$ and $P(\rho^1, \varepsilon)$ s. Therefore, a $C_{n-1}(\rho, \varepsilon)$ can be established.

3.5.2 $\exists \rho_j=1$, and ρ_1 and every ρ_i are external, where $2 \leq i, j \leq k$ and $i \neq j$

There is a bottleneck in this case for $k=3$; due to k is too small, the exchange space is not enough. In other words, $k=3$ is particular and $k \geq 4$ is general in this case. The following first considers building vertex-disjoint paths for $k=3$ and then concentrates on $k \geq 4$.

For $k=3$, let ρ has ρ_1, ρ_j and ρ_g . Vertex ρ has two ρ^1 s, a ρ^j , a ρ^g and $n-5$ ρ^1 s. Five constructions are listed below.

- i $P(\rho, \rho^1, \varepsilon)$ with $\rho_1^1 = g$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \varepsilon$.
- ii $P(\rho, \rho^1, \varepsilon)$ with $\rho_1^1 = j$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \rho^{1,j,g} \rightarrow \varepsilon$.
- iii $P(\rho, \rho^g, \varepsilon)$ with $\varepsilon_1^1 = \rho_1$ is constructed as $\rho \rightarrow \rho^g \rightarrow \varepsilon$.
- iv $P(\rho, \rho^j, \varepsilon)$ with $\varepsilon_1^1 = \rho_g$ is constructed as $\rho \rightarrow \rho^j \rightarrow \varepsilon$.
- v $P(\rho, \rho^1, \varepsilon)$ s are constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \varepsilon$. These $n-5$ $P(\rho, \rho^1, \varepsilon)$ s are vertex-disjoint by Lemma 3.

According to Lemma 3, each vertex in **v** is vertex-disjoint to **i**, **ii**, **iii** and **iv**. Path **i** is vertex-disjoint to **ii**, because **i** has a unique ρ^1 and every vertex in $P(\rho^{1,j}, \varepsilon)$ has fixed symbol j , but no such vertex in **ii**. Path **i** is vertex-disjoint to **iii** and **iv** by Lemma 1. Second, **ii** is vertex-disjoint to **iii** because each vertex in **iii** has ρ_1 , but no such vertex in **ii**. Path **ii** is vertex-disjoint to **iv** because each vertex in **ii** has symbol 1 and no ρ_1 , but each vertex in **iv** has either symbols 1 and ρ_1 or no symbol 1. Path **iii** is vertex-disjoint to **iv** because they have unique ε^1 s, and each vertex in **iii** has ρ_1 at position g , but no such vertex in **iv**. Therefore, a $C_{n-1}(\rho, \varepsilon)$ can be formed.

For $k \geq 4$, let symbol $\rho_1^1 = x$. For ease of description, all constructed paths are separated into six groups **i-vi**. Groups **i** and **ii** are $P(\rho, \rho^1, \varepsilon)$ s; groups **iii-v** are $P(\rho, \rho^i, \varepsilon)$ s, and group **vi** is $P(\rho, \rho^1, \varepsilon^{1'})$ s.

- i $P(\rho^1, \varepsilon^1)$ s for $2 \leq x \leq k-1$ with $\varepsilon_1^1 \neq \rho_1$ are constructed as $\rho^1 \rightarrow \rho^{1,x} \rightarrow \varepsilon^1$. These $k-2$ $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint by Lemma 1.
- ii $P(\rho, \rho^1, \varepsilon^{1'})$ for $x=k$ is constructed as $\rho \rightarrow \rho^{1,k} \rightarrow \rho^{1,k,1,j} \rightarrow \varepsilon^{1'}$.
- iii $P(\rho, \rho^j, \varepsilon^1)$ is constructed as $\rho \rightarrow \rho^j \rightarrow \varepsilon^1$.
- iv $P(\rho, \rho^i, \varepsilon^i)$ s, where $i \neq j \neq 1$, are constructed as $\rho \rightarrow \rho^i \rightarrow \rho^{i,1,j} \rightarrow \rho^{i,1,j,1} \rightarrow \varepsilon^i$. These $k-2$ $P(\rho^i, \varepsilon^i)$ s are vertex-disjoint by Lemma 2.

- v $P(\rho, \rho^{j \oplus 1}, \varepsilon^j)$ is constructed as $\rho \rightarrow \rho^{j \oplus 1} \rightarrow \varepsilon^j$.
- vi $P(\rho, \rho^1, \varepsilon^1)$ s are constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \varepsilon^1$. According to Lemma 3, the constructed $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint.

Group **ii** is vertex-disjoint to each path in **i** because each path in **i** and $P(\rho^1, \rho^{1,k,1,j})$ have a unique x , and each vertex of $P(\rho^{1,k,1,j}, \varepsilon^{j \oplus 1})$ has symbol 1 at position $j \oplus 1$, but no such vertex in **i**. Group **iii** is vertex-disjoint to each paths in **i** and **ii**, because every vertex in **iii** has ρ_1 , but no such vertex in **i** and **ii**. Each path in **iv** is vertex-disjoint to each path in **i** and **ii** because every vertex in **iv** has either ρ_1 or symbol 1 at position i , but no such vertex in **i** and **ii**. Group **iv** is vertex-disjoint to the path in **iii**, because every vertex in **iii** has ρ_1 at position j , but no such vertex in **iv**. Hence, paths in **i**, **ii**, **iii** and **iv** are vertex-disjoint.

The path in **v** is vertex-disjoint to each path in **i** and **ii** because vertex in **v** has symbol 1 and either ρ_1 or symbols $j \oplus 1$ and $j \oplus 2$, but each vertex with symbol 1 in **i** and **ii** neither has ρ_1 or symbols $j \oplus 1$ and $j \oplus 2$. The path in **v** is vertex-disjoint to the path in **iii**, because every vertex in **v** has symbol 1 at position j , but no such vertex in **iii**. The path in **v** is vertex-disjoint to each path in **iv**, because every vertex in **iv** has ρ_1 or symbol 1 at position i , but no such symbol at position i in each vertex of **v**. Each path in **vi** is vertex-disjoint to each path of **i-v** by Lemma 3. Therefore, paths in **i-vi** are vertex-disjoint, and a $C_{n-1}(\rho, \varepsilon)$ can be formed.

3.5.3 Symbols of ρ exist one misplaced and else are external

Sections 3.5.3.1 and 3.5.3.2 are adopted to describe constructions of $C_\omega(\rho, \varepsilon)$ s for ρ_1 is misplaced and not misplaced, respectively. Obviously, they also have a bottleneck when $k=3$, so $k=3$ is particular and $k \geq 4$ is general in each subsection.

3.5.3.1 Symbol ρ_1 is misplaced and every ρ_i are external, where $2 \leq i \leq k$

For $k=3$, let ρ has symbols ρ_1, ρ_j and ρ_g , where $\rho_1=j$. Vertex ρ has two ρ^1 s, a ρ^g , a ρ^j and $n-5$ ρ^1 s. Five constructions are listed below.

- i $P(\rho, \rho^1, \varepsilon)$ with $\rho_1^1 = 1$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \varepsilon$.
- ii $P(\rho, \rho^1, \varepsilon^1)$ with $\rho_1^1 = g$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,g} \rightarrow \varepsilon^1$.
- iii $P(\rho, \rho^g, \varepsilon^1)$ with $\varepsilon_1^1 = \rho_1$ is constructed as $\rho \rightarrow \rho^g \rightarrow \rho^{g,1} \rightarrow \rho^{g,1,g} \rightarrow \varepsilon^1$.
- iv $P(\rho, \rho^j, \varepsilon^1)$ with $\varepsilon_1^1 = \rho_g$ is constructed as $\rho \rightarrow \rho^j \rightarrow \rho^{j,g} \rightarrow \varepsilon^1$.
- v $P(\rho, \rho^1, \varepsilon^1)$ s are constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \varepsilon^1$. According to Lemma 3, the constructed $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint.

According to Lemma 3, each vertex in $\mathbf{v}$ is vertex-disjoint to $\mathbf{i}$, $\mathbf{ii}$, $\mathbf{iii}$ and $\mathbf{iv}$. Path in $\mathbf{i}$ is vertex-disjoint to each vertex in $\mathbf{ii}$, $\mathbf{iii}$ and $\mathbf{iv}$, because $\mathbf{i}$ has a unique ρ^1 and every vertex in $P(\rho^{1,j}, \varepsilon)$ has symbol 1 at position j , but no such vertex in $\mathbf{ii}$, $\mathbf{iii}$ and $\mathbf{iv}$. Path in $\mathbf{iii}$ is vertex-disjoint to each vertex in $\mathbf{ii}$ and $\mathbf{iv}$, because each vertex in $\mathbf{iii}$ has j or symbol 1 at position g . Path in $\mathbf{ii}$ is vertex-disjoint to $\mathbf{iv}$ because they have a unique ε^1 , and each vertex in $\mathbf{iv}$ has fixed symbol j , but no such vertex in $\mathbf{ii}$. Therefore a $C_{n-1}(\rho, \varepsilon)$ can be formed.

For $k \geq 4$, let $\rho_1 = j$ and $\rho_1^1 = x$. For clarity, all constructed paths are divided in six groups $\mathbf{i}$ – $\mathbf{vi}$. Groups $\mathbf{i}$ – $\mathbf{iii}$ are $P(\rho, \rho^1, \varepsilon)$ s; groups $\mathbf{iv}$ and $\mathbf{v}$ are $P(\rho, \rho^j, \varepsilon)$ s, and group $\mathbf{vi}$ is $P(\rho, \rho^1, \varepsilon)$ s.

- $\mathbf{i}$ $P(\rho^1, \varepsilon^j, \varepsilon)$ for $x=1$ is constructed as $\rho^1 \rightarrow \rho^{1,j} \rightarrow \langle j \oplus 1, \dots, k, 2, \dots, j \rangle \varepsilon$.
- $\mathbf{ii}$ $P(\rho^1, \varepsilon^1, \varepsilon)$ for $x=j \oplus 1$ and $\varepsilon_1^1 = \rho_j$ is constructed as $\rho^1 \rightarrow \rho^{1,j \oplus 1} \rightarrow \langle j \oplus 1, \dots, k, 2, \dots, j \oplus 2, j \rangle \varepsilon^1$.
- $\mathbf{iii}$ $P(\rho^1, \varepsilon^{x \oplus 1}, \varepsilon)$ s for $2 \leq x \leq k$ and $x \neq j \oplus 1$ are constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,x} \rightarrow \rho^{1,x,1} \rightarrow \rho^{1,x,1, x \oplus 1} \rightarrow \langle x \oplus 2, \dots, k, 2, \dots, x \oplus 1, x \oplus 1 \rangle \varepsilon$ with $\rho_1^{1,x,1} = 1$. These $k-2$ paths are vertex-disjoint, because each path has a unique symbol x and each vertex in it has either symbol 1 at position $x \oplus 1$ or symbol x .
- $\mathbf{iv}$ $P(\rho^j, \varepsilon^{j \oplus 1}, \varepsilon)$ is constructed as $\rho \rightarrow \rho^j \rightarrow \rho^{j,1} \rightarrow \rho^{j,1,j \oplus 1} \rightarrow \langle j \oplus 2, \dots, k, 2, \dots, j \oplus 1, j \oplus 1 \rangle \varepsilon$ with $\rho_1^{j,1} = 1$.
- $\mathbf{v}$ $P(\rho^i, \varepsilon^1)$ s with $\varepsilon_1^1 \neq \rho_j$ are constructed as $\rho \rightarrow \rho^i \rightarrow \langle i, j \oplus 1, \dots, k, 2, \dots, i \oplus 1 \rangle \varepsilon^1$. These $k-2$ paths are vertex-disjoint because each vertex in each path without fixed j has symbol ρ_1 or i at different position i , and subpaths with fixed j are vertex-disjoint by Lemma 1.
- $\mathbf{vi}$ $n-2k+1$ vertex-disjoint $P(\rho^1, \varepsilon^1)$ s are constructed and proved as that in Case 1.1.

Group $\mathbf{vi}$ is vertex-disjoint to each path in $\mathbf{i}$ – $\mathbf{v}$ by Lemma 3. The path in $\mathbf{i}$ is vertex-disjoint to each path in $\mathbf{ii}$ – $\mathbf{v}$, because $\mathbf{i}$ has a unique ρ^1 and every vertex in $P(\rho^{1,j}, \varepsilon)$ has symbol 1 at position j , but no such vertex in $\mathbf{ii}$ – $\mathbf{v}$.

The path in $\mathbf{ii}$ is vertex-disjoint to each paths in $\mathbf{iv}$ and $\mathbf{v}$, because vertex in $\mathbf{iv}$ and $\mathbf{v}$ has symbol j ; each vertex in $\mathbf{ii}$ excluding ε^1 and $\varepsilon^{1,j}$ does not have symbol j , and vertices ε^1 and $\varepsilon^{1,j}$ in $\mathbf{ii}$ are composed of symbols 2 to k and a ρ_j , but no such vertex in $\mathbf{iv}$ and $\mathbf{v}$. The path in $\mathbf{ii}$ is vertex-disjoint to the path in $\mathbf{iii}$, because each of them has a unique ρ^1 , each vertex in $\mathbf{ii}$ does not has symbol 1 and each vertex in $P(\rho^{1,j \oplus 1}, \varepsilon^1)$ has a fixed symbol $j \oplus 1$; each vertex in $\mathbf{iii}$ excluding $\rho^{1,x}$ and $\rho^{1,x,1}$ has symbol 1, and symbol $j \oplus 1$ is not fixed in $\rho^{1,x}$ and $\rho^{1,x,1}$.

Group $\mathbf{iii}$ is vertex-disjoint to the path in $\mathbf{iv}$, because every vertex in $\mathbf{iv}$ has fixed symbol j and it has either no symbol 1 or symbol 1 at position $j \oplus 1$, but each vertex in $\mathbf{iii}$ with fixed symbol j also has symbol 1 at position $i \oplus 1$. Group $\mathbf{iii}$ is vertex-disjoint to the path in $\mathbf{v}$, because every vertex in $\mathbf{iii}$ has either

symbol 1 or no symbol j , but each vertex in $\mathbf{v}$ has symbol j and no symbol 1. The path in $\mathbf{iv}$ is vertex-disjoint to the path in $\mathbf{v}$, because ρ^j and ρ^i in $\mathbf{iv}$ and $\mathbf{v}$ are unique, and every vertex in $P(\rho^{i,1}, \varepsilon)$ of $\mathbf{iv}$ has fixed symbol j and symbol 1, but no vertex has symbol 1 in $\mathbf{v}$. Therefore, paths in $\mathbf{i}$, $\mathbf{ii}$, $\mathbf{iii}$, $\mathbf{iv}$, $\mathbf{v}$ and $\mathbf{vi}$ are vertex-disjoint, and a $C_{n-1}(\rho, \varepsilon)$ can be formed.

3.5.3.2 $\exists \rho_j$ is misplaced, ρ_1 and every ρ_i are external for $2 \leq i, j \leq k$ and $i \neq j$

For $k=3$, misplaced symbol $\rho_j=i$. Vertex ρ has two ρ^1 s, a ρ^j , a ρ^i and $n-5$ ρ^1 s. Five constructions are listed below:

- $\mathbf{i}$ $P(\rho, \rho^1, \varepsilon)$ with $\rho_1^1=1$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \langle i, j \rangle \varepsilon$.
- $\mathbf{ii}$ $P(\rho, \rho^1, \varepsilon)$ with $\rho_1^1=j$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \langle j \rangle \varepsilon^1 \rightarrow \varepsilon$.
- $\mathbf{iii}$ $P(\rho, \rho^j, \varepsilon)$ is constructed as $\rho \rightarrow \rho^j \rightarrow \langle i, j \rangle \varepsilon^1 \rightarrow \varepsilon$.
- $\mathbf{iv}$ $P(\rho, \rho^i, \varepsilon)$ is constructed as $\rho \rightarrow \rho^i \rightarrow \rho^{i,1} \rightarrow \rho^{i,1,i} \rightarrow \langle j, j \rangle \varepsilon$ with $\rho_1^{i,1}=1$.
- $\mathbf{v}$ $P(\rho, \rho^1, \varepsilon)$ s are constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \langle i, j \rangle \varepsilon^1 \rightarrow \varepsilon$.

According to Lemma 3, each vertex in $\mathbf{v}$ is vertex-disjoint to $\mathbf{i}$, $\mathbf{ii}$, $\mathbf{iii}$ and $\mathbf{iv}$. First, $\mathbf{i}$ is vertex-disjoint to $\mathbf{ii}$, $\mathbf{iii}$ and $\mathbf{iv}$, because $\mathbf{i}$ has a unique ρ^1 and every vertex in $P(\rho^{1,j}, \varepsilon)$ has symbol 1 at position j , but no such vertex in $\mathbf{ii}$, $\mathbf{iii}$ and $\mathbf{iv}$. Second, $\mathbf{ii}$ is vertex-disjoint to $\mathbf{iii}$ and $\mathbf{iv}$, because each vertex in $\mathbf{iii}$ has ρ_1 and each vertex $\mathbf{iv}$ has ρ_1 or symbol 1, but each vertex in $\mathbf{ii}$ has neither ρ_1 or symbol 1. Third, $\mathbf{iii}$ is vertex-disjoint to $\mathbf{iv}$ because vertices ρ^j and $\rho^{i,1}$ in $\mathbf{iv}$ have i at position j and other vertices in $\mathbf{iv}$ have symbol 1, but each vertex in $\mathbf{iii}$ has neither symbol 1 or i at position j . Therefore a $C_{n-1}(\rho, \varepsilon)$ can thus be formed.

For $k \geq 4$, paths $P(\rho^1, \varepsilon^1)$ s can be similarly constructed as that in Case 1.2, but the constructions of $P(\rho^1, \varepsilon)$ s and $P(\rho^j, \varepsilon)$ s should be reconsidered. Due to $C_\omega^*(\rho, \varepsilon)$ may not be unique, only one $C_\omega^*(\rho, \varepsilon)$ should be built. In other words, to build a $C_\omega^*(\rho, \varepsilon)$, the major problem is to make all constructed $P(\rho^\alpha, \varepsilon^\beta)$ s disjoint and short enough, where $\alpha \in \{1, i\}$ and $\beta \in \{i, 1\}$. Hence, before constructing each path of $P(\rho^1, \varepsilon)$ s and $P(\rho^j, \varepsilon)$ s, a pair of ρ^α and ε^β should be first determined.

For avoiding $l(P(\rho, \rho^1, \varepsilon^j, \varepsilon)) > d(S_{n,k})$, each pair of ρ^1 and ε^j should be carefully chosen. Significantly, each ρ^1 has its proper ε^j s and only one of the proper ε^j s will be chosen for the ρ^1 . The path construction process is that all proper ε^j s are first selected for each ρ^1 , then choose one of the proper ε^j s for each ρ^1 , and finally build $P(\rho, \rho^1, \varepsilon^j, \varepsilon)$ s. The detailed process is listed below.

Assign each ρ^1 a destination set $\{2, 3, \dots, k\}$ to choose proper ε^j s. Let x and y represent ρ_1^1 and ρ_j , respectively. When $x \neq j$ and $2 \leq x \leq k$, each ρ^1 should avoid to choose ε^j or ε^x . When $x=j$, the ρ^1 should

avoid to choose ε^j and ε^y . Hence, each ρ^1 deletes some symbols from its destination set and then every ρ^1 has a destination set $\{2, 3, \dots, k\} - \{j, x\}$ or $\{2, 3, \dots, k\} - \{j, y\}$ for $x \neq j$ or $x = j$, respectively. Additionally, reassign the ρ^1 a destination set $\{j\}$ for $x=1$. To choose a proper ε^i , each ρ^1 has to choose one symbol from its destination set. First, create a removed set to record the symbols have been chosen and a source set $\{2, 3, \dots, k\} - \{\rho_j\}$. Second, to determine every pair of ρ^1 and ε^i , three necessary steps are stated below.

Step 1: Choose the ρ^1 with x is the smallest symbol in the source set and the smallest symbol s in its destination set. Remove x from the source set and then add s into the removed set.

Step 2: Select one ρ^1 with x is in the source set such that its destination set has the maximum number of symbols including the symbol just added in the removed set. Remove all symbols in the removed set from the destination set of the ρ^1 . Choose the smallest symbol s in its destination set. Remove the current x from the source set and then add s into the removed set.

Step 3: Repeat Step 2 until the source set is empty.

In order to simplify the description, constructions of $P(\rho, \rho^1, \varepsilon^i, \varepsilon)$ s are divided into three groups **i**, **ii** and **iii**.

- i** $P(\rho, \rho^1, \varepsilon^i, \varepsilon)$ with $x=1$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow_{< y, j \oplus 1, \dots, k, 2, \dots, j > \varepsilon}$.
- ii** $P(\rho, \rho^1, \varepsilon^i, \varepsilon)$ with $x=j$ is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j} \rightarrow \rho^{1,j,y} \rightarrow \rho^{1,j,y,1} \rightarrow \rho^{1,j,y,1,s} \rightarrow_{< s \oplus 1, \dots, k, 2, \dots, s > \varepsilon}$ with $\rho^{1,j,y,1}=1$. Notably, j and y are not included in $< s \oplus 1, \dots, k, 2, \dots, s >$, because j and y are fixed symbols in $\rho^{1,j,y,1,s}$.
- iii** $P(\rho, \rho^1, \varepsilon^i, \varepsilon)$ s, where $2 \leq x \leq k$ and $x \neq j$, is constructed as $\rho \rightarrow \rho^1 \rightarrow \rho^{1,x} \rightarrow \rho^{1,x,1} \rightarrow \rho^{1,x,1,j} \rightarrow \rho^{1,x,1,j,y} \rightarrow \rho^{1,x,1,j,y,1} \rightarrow \rho^{1,x,1,j,y,1,s} \rightarrow_{< s \oplus 1, \dots, k, 2, \dots, s > \varepsilon}$ with $\rho^{1,x,1,j,y,1}=1$. Notably, x, j and y are not included in $< s \oplus 1, \dots, k, 2, \dots, s >$, because x, j and y are fixed symbols in $\rho^{1,x,1,j,y,1,s}$. Significantly, symbol x in every vertex of each $P(\rho^{1,x}, \varepsilon^i) = P(\rho^{1,x}, \varepsilon^s)$ is fixed and every vertex u in each $P(\rho^{1,x,1,j,y,1,s}, \varepsilon^i) = P(\rho^{1,x,1,j,y,1,s}, \varepsilon^s)$ has $u_s=1$. Since each $P(\rho^1, \varepsilon^i)$ has unique x and s , all $P(\rho^1, \varepsilon^i)$ s constructed in this group are vertex-disjoint.

The path of group **i** is vertex-disjoint to each path in **ii** and **iii**, because **i** has a unique ρ^1 , and every vertex in **i** after ρ^1 has 1 at position j , but no such vertex in both **ii** and **iii**. Similar to the principle described in **iii**, the $P(\rho^1, \varepsilon^i)$ in **ii** and each $P(\rho^1, \varepsilon^i)$ in **iii** are vertex disjoint, because they have different x and s . Therefore, all constructed $P(\rho^1, \varepsilon^i)$ s are vertex-disjoint.

For brevity, four groups **iv**–**vii** are used to describe the constructions of $P(\rho^i, \varepsilon^1)$ s and Path $P(\rho^j, \varepsilon^1)$.

iv Path $P(\rho^j, \varepsilon^1)$ with $\varepsilon_1^1 = \rho_1$ is constructed as

- $\rho^j \rightarrow \rho^{j,y} \rightarrow_{< j \oplus 1, \dots, k, 2, \dots, j > \varepsilon^1$.
- v** Path $P(\rho^{j \oplus 1}, \varepsilon^1)$ with $\varepsilon_1^1 = \rho_y$ is constructed as $\rho^{j \oplus 1} \rightarrow_{< j \oplus 1, \dots, k, 2, \dots, j, y > \varepsilon^1$.
- vi** Path $P(\rho^{j \oplus 1}, \varepsilon^1)$ with $\varepsilon_1^1 = \rho_{j \oplus 1}$ is constructed as $\rho^{j \oplus 1} \rightarrow_{< j, y, j \oplus 2, \dots, k, 2, \dots, j \oplus 1 > \varepsilon^1$.
- vii** Paths $P(\rho^h, \varepsilon^1)$ s with $\varepsilon_1^1 = \rho_{h \oplus 1}$, for $2 \leq h \leq k$ and $h \notin \{j, j \oplus 1, y \oplus 1\}$ are constructed as $\rho^h \rightarrow_{< j, y, h \oplus 2, \dots, k, 2, \dots, h \oplus 1 > \varepsilon^1$. Let u_h represent the first vertex, which has fixed symbols j and y , in each $P(\rho^h, \varepsilon^1)$. Apparently, each vertex in $P(u_h, \varepsilon^1)$ has fixed symbols j and y and each vertex in each $P(\rho^h, u_h)$ has ρ_1 at position h . Because each $P(u_h, \varepsilon^1)$ is built according to a unique $< h \oplus 2, \dots, k, 2, \dots, h \oplus 1 >$, by Lemma 1 every two $P(u_h, \varepsilon^1)$ s are vertex disjoint. Moreover, since each $P(\rho^h, \varepsilon^1)$ has a unique h and, $P(\rho^h, \varepsilon^1)$ s constructed are vertex-disjoint.

Path **iv** is vertex-disjoint to each paths in **v**, **vi** and **vii**, because each vertex in **iv** has ρ_1 at position j but no such vertex in **v**, **vi** and **vii**. Path in **v** is vertex-disjoint to each path in **vi** and **vii** by Lemma 1. Similar to the principle described in **iii**, the path in **vi** is vertex-disjoint to each path in **vii**. Thus, $P(\rho^j, \varepsilon)$ s are vertex-disjoint. Notably, vertices in each $P(\rho^1, \varepsilon^i)$ do not have ρ_1 and some of them have 1, and vertices in each $P(\rho^1, \varepsilon^i)$ do not have 1 and some of them have ρ_1 . In other words, some vertices in both $P(\rho^1, \varepsilon^i)$ s and $P(\rho^i, \varepsilon^1)$ s, each of them is denoted by μ , do not have 1 and ρ_1 . Some of μ s in each $P(\rho^1, \varepsilon^i)$ have unfixed j or y ; some of μ s in each $P(\rho^1, \varepsilon^i)$ have fixed j and y , and every μ in each $P(\rho^i, \varepsilon^1)$ have fixed j, y . Those μ s in each $P(\rho^1, \varepsilon^i)$ with fixed j and y have ρ_y at position 1 and every μ in each $P(\rho^i, \varepsilon^1)$ does not have ρ_y . Based on what mention above, each $P(\rho^1, \varepsilon^i)$ is vertex-disjoint to each $P(\rho^i, \varepsilon^1)$. According to Lemma 3 each $P(\rho^1, \varepsilon^1)$ is vertex-disjoint to each $P(\rho^1, \varepsilon^i)$ and each $P(\rho^i, \varepsilon^1)$. Therefore, $k-1$ $P(\rho^1, \varepsilon^i)$ s, $k-1$ $P(\rho^i, \varepsilon^1)$ s, and $n-2k+1$ $P(\rho^1, \varepsilon^1)$ s are vertex-disjoint and a $C_{n-1}(\rho, \varepsilon)$ can thus be formed.

3.6 Case 2.3: $\rho_1 \neq 1$, and ρ has $k-2$ external symbols

Since there are two indefinite symbols in ρ as shown in Table 1, they can be that one is a misplaced symbol and the other is symbol 1, misplaced or fixed symbol. The case, which has one misplaced and the other is misplaced or fixed symbol, can be dominated by 3.5.3. Hence, only the case, which has one misplaced symbol and 1, should be discussed. Two cases should be further discussed as follows.

3.6.1 $\rho_1 = g$ is misplaced; $\exists \rho_g = 1$, and every ρ_i is external, where $2 \leq i, j, g \leq k$ and $i \notin \{j, g\}$

For showing the paths in different constructions are vertex-disjoint clearly, constructed paths are

divided in nine groups **i–ix**: Group **i** is $P(\rho^1, \varepsilon)$ s; groups **ii–vii** are $P(\rho^i, \varepsilon)$ s, and groups **viii** and **ix** are $P(\rho^{1'}, \varepsilon)$ s. For $1 \leq \omega \leq 2k-3$, if $g \neq j$, then constructions follow **i–iii** and **vi**; otherwise, constructions follow **i** and **iv–vi**. For $2k-2 \leq \omega \leq n-1$, if $g \neq j$ then constructions follow **i**, **iii**, and **vi–ix**; otherwise, constructions follow **i** and **vi–ix**. For ease of use, let h represent the smallest symbol of i and l represent an outer external symbol.

- i** Totally, $k-3$ $P(\rho^1, \varepsilon^{1'})$ s with $\rho_1^1 = x$, for $2 \leq x \leq k$, $x \notin \{\rho_1, j \oplus 1\}$, can be built as $\rho^1 \rightarrow \rho^{1,x} \rightarrow \langle x \oplus 1, \dots, k, 2, \dots, x \oplus 1 \rangle \varepsilon^{1'}$. One $P(\rho^1, \varepsilon^{1'})$ with $\rho_1^1 = j \oplus 1$ can be built in three cases depending on ρ_1 . Case a. ($\rho_1 = j \oplus 1$): $\rho^1 \rightarrow \rho^{1,j \oplus 1} \rightarrow \langle j \oplus 2, \dots, k, 2, \dots, j \oplus 1 \rangle \varepsilon^{1'}$. Case b. ($\rho_1 = j$): $\rho^1 \rightarrow \rho^{1,j \oplus 1} \rightarrow \langle j, j \oplus 2, \dots, k, 2, \dots, j \oplus 1 \rangle \varepsilon^{1'}$. Case c. ($\rho_1 \notin \{j, j \oplus 1\}$): $\rho^1 \rightarrow \rho^{1,j \oplus 1} \rightarrow \langle j \oplus 1, j, j \oplus 2, \dots, k, 2, \dots, j \oplus 2 \rangle \varepsilon^{1'}$. By Lemma 1, $k-2$ $P(\rho^1, \varepsilon^{1'})$ s are vertex-disjoint.
- ii** Path $P(\rho^g, \varepsilon^j, \varepsilon)$ is constructed as $\rho^g \rightarrow \langle g \oplus 1, \dots, k, 2, \dots, g \oplus 1, j \rangle \varepsilon$.
- iii** Path $P(\rho^j, \varepsilon^g, \varepsilon)$ is constructed as $\rho^j \rightarrow \rho^{j,g} \rightarrow \langle g \oplus 1, \dots, k, 2, \dots, g \oplus 1, j \rangle \varepsilon$.
- iv** Path $P(\rho^j, \varepsilon^h, \varepsilon)$ is constructed as $\rho^j \rightarrow \rho^{j,h} \rightarrow \langle h \oplus 1, \dots, k, 2, \dots, h \oplus 1, j \rangle \varepsilon$.
- v** Path $P(\rho^h, \varepsilon^j, \varepsilon)$ is constructed as $\rho^h \rightarrow \langle j \oplus 1, \dots, k, 2, \dots, j \oplus 1, h \rangle \varepsilon$.
- vi** Paths $P(\rho^i, \varepsilon^j, \varepsilon)$ s are constructed as $\rho^i \rightarrow \langle j \oplus 1, \dots, k, 2, \dots, j \oplus 1, i \rangle \varepsilon$. Notably, correcting sequence $\langle j \oplus 1, \dots, k, 2, \dots, i \rangle$ excludes j and g , and if $1 \leq \omega \leq 2k-3$ and $g = j$ then $i \neq h$ and only $k-3$ $P(\rho^i, \varepsilon^j)$ s can still be built. According to Lemma 2, these $k-3$ $P(\rho^i, \varepsilon^j, \varepsilon)$ s are vertex-disjoint. Similarly, for $2k-2 \leq \omega \leq n-1$ and $g = j$, the constructed $k-2$ $P(\rho^i, \varepsilon^j, \varepsilon)$ s are also vertex-disjoint.
- vii** Path $P(\rho^g, \varepsilon^{1'}, \varepsilon)$ with $\varepsilon_1^{1'} = l$ is constructed as $\rho^g \rightarrow \rho^{g,1'} \rightarrow \rho^{g,1',g \oplus 1} \rightarrow \langle g \oplus 1, \dots, k, 2, \dots, g \oplus 1 \rangle \varepsilon^{1'}$ with $\rho_1^{g,1'} = l$.
- viii** Path $P(\rho^{1'}, \varepsilon^j, \varepsilon)$ with $\rho_1^{1'} = l$ is constructed as $\rho^{1'} \rightarrow \rho^{1',j \oplus 1} \rightarrow \langle j \oplus 1, \dots, k, 2, \dots, j \oplus 1 \rangle \varepsilon$.
- ix** Paths $P(\rho^{1'}, \varepsilon^{1'}, \varepsilon)$ s with $\rho_1^{1'} \neq l$ and $\varepsilon_1^{1'} \neq l$ can be similarly constructed as Case 3.5.2. With the aid of Lemma 3, the constructed $n-2k+1$ $P(\rho^{1'}, \varepsilon^{1'})$ s are vertex-disjoint.

Each path in **ix** is vertex-disjoint to each path in other groups by Lemma 3. Groups **i** and **ii** are vertex-disjoint by Lemma 1. Group **ii** is vertex-disjoint to each path in **iii** and **vi** because every vertex in **ii** has 1 at position j and fixed symbol g , but no such vertex in **iii** and **vi**. Each path in **i** is vertex-disjoint to each path in **iii** and **vi**, due to every vertex in **iii** and **vi** has unfixed ρ_1 or 1 at position i , but no such vertex in **i**. Path **iii** is vertex-disjoint to **vi** by Lemma 2. Group **iv** is vertex-disjoint to each path in **i**, **v** and **vi** because every vertex in **iv** has 1 at position y , but no such vertex in **i**, **v** and **vi**. Group **v** is vertex-disjoint to each path in **i** and **vi** because every vertex in **v** has ρ_1 at position y , but no such vertex in **i** and **vi**.

Groups **vii** and **viii** are vertex-disjoint, because $\rho_1^{g,1'} = l$, $\rho_g^{g,1'} = g$, $\rho_j^{g,1'} = 1$ and every $\rho_i^{g,1'}$ is external, but no such vertex in **viii**; ρ^g and $\varepsilon^{1'}$ are unique, and every vertex in $P(\rho^{g,1',g \oplus 1}, \varepsilon^{1',g \oplus 1})$ has l at position $g \oplus 1$, but no such vertex in **viii**. Groups **vii** and **viii** are also vertex-disjoint to each path in **i**, **ii**, **iii** and **vi** by Lemma 3. Therefore, groups **i–ix** are vertex-disjoint and a $C_{n-1}(\rho, \varepsilon)$ can be formed.

3.6.2 ρ_1 is external; $\exists \rho \neq 1$ and $\exists \rho_g$ is misplaced, $j \neq g$, and every ρ_i is external, where $2 \leq i, j, g \leq k$ and $i \notin \{j, g\}$

Since the numbers of ρ^1 and ρ^j are $k-2$ and $k-1$, respectively, $k-2$ vertex-disjoint $P(\rho^1, \varepsilon^{1'})$ s and $k-1$ vertex-disjoint $P(\rho^j, \varepsilon^j)$ s can be similarly constructed and proved as groups **i–iv** in Section 3.6.1. Moreover, $n-2k+2$ vertex-disjoint $P(\rho^{1'}, \varepsilon^{1'})$ s can be similarly constructed and proved as that in Case 1.2. According to Lemma 3, each $P(\rho^{1'}, \varepsilon^{1'})$ is vertex-disjoint to each $P(\rho^1, \varepsilon^{1'})$ and each $P(\rho^j, \varepsilon^j)$. That each $P(\rho^1, \varepsilon^{1'})$ and each $P(\rho^j, \varepsilon^j)$ are vertex-disjoint can be proved as that in Case 3.6.1. Therefore, a $C_{n-1}(\rho, \varepsilon)$ can be formed.

4 ω -Wide Diameters

This section shows every $C_\omega(\rho, \varepsilon)$ constructed in Section 3 is a $C_\omega^*(\rho, \varepsilon)$ for $1 \leq \omega \leq n-1$, computes $d_\omega(\rho, \varepsilon)$ for each selected vertex ρ , and determines $d_\omega(S_{n,k})$ for $2 \leq k \leq \lfloor n/2 \rfloor$. To compute $d_\omega(S_{n,k})$, analyze every constructed $C_\omega(\rho, \varepsilon)$, show every constructed $C_\omega(\rho, \varepsilon)$ is a $C_\omega^*(\rho, \varepsilon)$, and then calculate each $d_\omega(\rho, \varepsilon)$. Lemma 4 and Lemma 5 compute the lengths of some paths.

Lemma 4. If $\rho = \rho_1 \rho_2 \dots \rho_k$, ρ_1 is 1 or an external symbol and every ρ_i is external, $2 \leq i \leq k$, then $d(\rho, \varepsilon) = d(S_{n,k})$.

Proof. Constructing a shortest path is fixing all symbols without redundant steps. Based on the routing rules provided by Chiang and Chen [11], a $P^*(\rho, \varepsilon)$ can be constructed according to one of $\langle 2, 3, \dots, k, 1 \rangle$, $\langle 3, 4, \dots, k, 2, 1 \rangle$, $\dots$, and $\langle k, 2, \dots, k-1, 1 \rangle$. Hence, $d(\rho, \varepsilon) = l(P^*(\rho, \varepsilon)) = d(S_{n,k})$. $\square$

Lemma 5. If $\rho^{1'}$ is composed of k external symbols and $\rho_1^{1'}$ is outer external, then the length of vertex-disjoint $P(\rho, \rho^{1'}, \varepsilon)$ s is $d(S_{n,k})+2$.

Proof. Obviously, $\rho^{1',1}$ ($\rho^{1',1}$) is the same as a ρ^1 ($\rho^{1'}$) and $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1',1} \rightarrow \rho$ ($\rho \rightarrow \rho^{1'} \rightarrow \rho^{1',1} \rightarrow \rho$) forms a cycle. Hence, $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1',i}$ for $2 \leq i \leq k$ is the only way to construct paths from vertex ρ passing through $\rho^{1'}$ to ε and it takes 2 steps from ρ to $\rho^{1',i}$. Additionally, $\rho^{1',i}$ is also composed of k external symbols, according to Lemma 4, $l(P(\rho^{1',i}, \varepsilon))$ is $d(S_{n,k})$. Therefore, the length of vertex-disjoint $P(\rho, \rho^{1'}, \varepsilon)$ s is $d(S_{n,k})+2$. $\square$

The lengths of the paths constructed in Section 3 are determined case by case as following.

4.1 Case 1.1: $\rho_1=1$ and every ρ_i is external for $2 \leq i \leq k$

$l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ is $2k-1$ because fixing every desired symbol takes 2 and $k-1$ desired symbols in the corresponding sequence takes $2k-2$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. $l(P(\rho, \rho^i, \varepsilon^i, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^i$ takes 1 and fixing $k-1$ desired symbols in the corresponding sequence takes $2k-2$. $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ is $2k+1$ because $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1'j}$ takes 2; fixing desired symbols 2 to k takes $2k-2$, and fixing symbol 1 takes 1. Thus, $k-1$ $P(\rho, \rho^1, \varepsilon^1, \varepsilon)$ s and $k-1$ $P(\rho, \rho^i, \varepsilon^i, \varepsilon)$ s have length $2k-1$ which is the same as the trivial lower bound $d(S_{n,k})$. According to Lemma 5, $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ should be $2k+1$, which is $d(S_{n,k})+2$. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq 2k-2$ and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+2$ if $2k-1 \leq \omega \leq n-1$.

4.2 Case 1.2: $\rho_1=1$, $\exists \rho_j$ is misplaced and every ρ_i is external for $2 \leq i, j \leq k$ and $i \neq j$

Referring to Section 4.1, the length of $k-2$ $P(\rho, \rho^1, \varepsilon^1, \varepsilon)$ s and $k-1$ $P(\rho, \rho^i, \varepsilon^i, \varepsilon)$ s is $2k-1$ which is the same as the trivial lower bound $d(S_{n,k})$. $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ is $2k$ because $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1'j}$ takes 2; fixing misplaced symbol ρ_j takes 1; fixing desired symbols 2 to k excluding ρ_j takes $2k-4$, and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1. Referring to the proof of Lemma 5, the minimum $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ is $d(S_{n,k})+1$ because a misplaced symbol is substituted for one of desired symbols in the correcting sequence and fixing it just take 1 step. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq 2k-3$ and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $2k-2 \leq \omega \leq n-1$.

4.3 Case 1.3: $\rho_1=1$, $\exists \rho_j$ and $\rho_g, j \neq g$, are misplaced and every ρ_i is external for $2 \leq i, j, g \leq k$ and $i \notin \{j, g\}$

Referring to Section 4.1, the length of $k-3$ $P(\rho, \rho^1, \varepsilon^1, \varepsilon)$ s and $k-1$ $P(\rho, \rho^i, \varepsilon^i, \varepsilon)$ s is $2k-1$ which is the same as the trivial lower bound $d(S_{n,k})$. To compute $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$, three cases depended on ρ_j and ρ_g as described in Section 3.3 should be considered as follows. In Case a, $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ is $2k$ because ρ to $\rho^{1'j,g,j,i}$ takes 5; fixing $k-3$ desired symbols takes $2k-6$, and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1. In Case b, $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1'j}$ takes 2; fixing 2 misplaced symbols ρ_g, ρ_j and $k-3$ desired symbols takes $2+2k-6$, and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1. In Case c, $l(P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^{1'} \rightarrow \rho^{1'j}$ takes 2; fixing 2 misplaced symbols ρ_g, ρ_j and $k-3$ desired symbols takes $2+2k-6$, and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1. Thus, $k-3$ $P(\rho, \rho^1, \varepsilon^1, \varepsilon)$ s and $k-1$ $P(\rho, \rho^i, \varepsilon^i, \varepsilon)$ s have length $2k-1$ which is the same as the trivial lower bound $d(S_{n,k})$. Restated, $P(\rho, \rho^{1'}, \varepsilon^{1'}, \varepsilon)$ in Case a cannot be shortened because two misplaced

symbols in the correcting sequence takes at least 3 steps (one misplaced symbol takes 2 steps and the other takes 1 step); $k-3$ desired symbols take at least $2k-6$ steps even without redundant steps, and fixing symbol 1 takes at least 1 step. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})+1$ if $2k-3 \leq \omega \leq n-1$, $\rho_j=g$ and $\rho_g=j$. Otherwise, $d_\omega(\rho, \varepsilon) = d(S_{n,k})$.

4.4 Case 2.1: ρ_1 and every ρ_i are external for $2 \leq i \leq k$

Because lengths of $k-1$ $P(\rho, \rho^i, \varepsilon)$ s and one $P(\rho, \rho^1, \varepsilon)$ with $\rho_1=1$ are at least $d(S_{n,k})+1$ by Lemma 4 and lengths of $n-2k$ $P(\rho, \rho^{1'}, \varepsilon)$ s are at least $d(S_{n,k})+2$ by Lemma 5, at most $k-1$ $P(\rho, \rho^1, \varepsilon)$ s with $\rho_1 \neq 1$ of length $d(S_{n,k})$ can be constructed.

For $1 \leq \omega \leq k-1$, referring to Section 4.1, the length of $k-1$ $P(\rho, \rho^1, \varepsilon)$ s with $\rho_1 \neq 1$ is exactly $d(S_{n,k})$. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq k-1$.

For $k \leq \omega \leq n-1$, when $\omega \geq 2k$ there exists a $P(\rho, \rho^{1'}, \varepsilon)$ whose length is at least $d(S_{n,k})+2$. Hence, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})+2$ if $\omega \geq 2k$. When $k \leq \omega \leq 2k-1$, there exists a $P(\rho, \rho^i, \varepsilon)$ or a $P(\rho, \rho^1, \varepsilon)$ with $\rho_1=1$ whose length is at least $d(S_{n,k})+1$. Significantly, if $l(C_\omega(\rho, \varepsilon)) \leq d(S_{n,k})+1$ then the $C_\omega(\rho, \varepsilon)$ is a $C_\omega^*(\rho, \varepsilon)$. Let $x=\rho_1^1$, analyzing lengths of vertex-disjoint paths are divided into four groups as follows.

- i For $3 \leq x \leq k-2$, $l(P(\rho, \rho^1, \varepsilon))$ is $2k$ because ρ to $\rho^{1,x,1,x \oplus 1}$ take 4 and fixing $k-2$ desired symbols takes $2k-4$. For $x=k \oplus 1$, $l(P(\rho, \rho^1, \varepsilon))$ is $2k$ because ρ to $\rho^{1,k \oplus 1,1,k}$ take 4 and fixing $k-2$ desired symbols takes $2k-4$. For $x=k$, $l(P(\rho, \rho^1, \varepsilon))$ is $2k$ because ρ to $\rho^{1,k,1,2}$ take 4 and fixing $k-2$ desired symbols takes $2k-4$. For $x=1$, $l(P(\rho, \rho^1, \varepsilon))$ is $2k$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,3}$ takes 2 and fixing $k-1$ desired symbols takes $2k-2$. Thus, $k-1$ $P(\rho, \rho^1, \varepsilon)$ s whose lengths are $d(S_{n,k})+1$.
- ii $l(P(\rho, \rho^1, \varepsilon))$ with $x=2$ is $2k-1$ because fixing $k-1$ desired symbols takes $2k-2$ and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1.
- iii $l(P(\rho, \rho^2, \varepsilon))$ is $2k$ because $\rho \rightarrow \rho^2$ takes 1; fixing $k-1$ desired symbols takes $2k-2$, and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1.
- iv $l(P(\rho, \rho^h, \varepsilon^1))$, where $3 \leq h \leq k$, is $2k$ because $\rho \rightarrow \rho^h$ take 1; fixing $k-1$ desired symbols takes $2k-2$, and $\varepsilon^{1'} \rightarrow \varepsilon$ takes 1.

Since lengths of vertex-disjoint paths in groups i–iv are no more than $2k=d(S_{n,k})+1$, $C_\omega^*(\rho, \varepsilon)$ can thus be formed for $k \leq \omega \leq 2k-1$. Consequently, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq k-1$; $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $k \leq \omega \leq 2k-1$, and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+2$ if $2k \leq \omega \leq n-1$.

4.5 Case 2.2: $\rho_1 \neq 1$, and ρ has $k-1$ external symbols

Similar to Section 3.5, three cases should be considered as follows for computing the length of each path.

4.5.1 $\exists \rho_j$ is fixed, and ρ_1 and every ρ_i are external for $2 \leq i, j \leq k$ and $i \neq j$

There are one $P(\rho^1, \varepsilon^j)$, one $P(\rho^j, \varepsilon^1)$, $k-2$ $P(\rho^1, \varepsilon^1)$ s, and $k-2$ $P(\rho^j, \varepsilon^j)$ s to be analyzed their lengths. $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ with $\varepsilon_1^1 = \rho_{i \oplus 1}$ is $2k-3$ because fixing $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. $l(P(\rho, \rho^1, \varepsilon^j, \varepsilon))$ with $\rho_1^1 = 1$ is $2k$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1j}$ takes 2, and fixing $k-1$ desired symbols takes $2k-2$. $l(P(\rho, \rho^j, \varepsilon^1, \varepsilon))$ with $\varepsilon_1^1 = \rho_1$ is $2k$ because $\rho \rightarrow \rho^j$ takes 1; fixing $k-1$ desired symbols takes $2k-2$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. $l(P(\rho, \rho^j, \varepsilon^j, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^j$ takes 1; fixing $k-3$ desired symbols takes $2k-6$, and $\rho^s \rightarrow \rho^{s,1} \rightarrow \rho^{s,1,i} \rightarrow \rho^{s,1,i,1} \rightarrow \varepsilon$ takes 4. In total, there are $n-2k+1$ $P(\rho^1, \rho^{1,i}, \varepsilon^1)$ s with $\varepsilon_1^1 = \rho_1^1$. $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,i}$ takes 2; fixing $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.

Only two paths whose length are $d(S_{n,k})+1$ need to be further discussed. Considering the $P(\rho, \varepsilon^j, \varepsilon)$, when j is a fixed symbol in ρ but not a fixed symbol in ε^j , since $P^*(\rho, \varepsilon^j)$ should encloses ε , the $P(\rho, \varepsilon^j)$ should not be shortest for excluding ε . Thus, $l(P(\rho, \varepsilon^j, \varepsilon))$ is at least $2k=d(S_{n,k})+1$. Considering the $P(\rho, \rho^j, \varepsilon)$, when j is not a fixed symbol in ρ^j , since $P^*(\rho^j, \varepsilon)$ should encloses ρ , the $P(\rho^j, \varepsilon)$ should not be shortest for excluding ρ . Thus, $l(P(\rho, \rho^j, \varepsilon))$ is at least $2k=d(S_{n,k})+1$. Consequently, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq n-3$, and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $n-2 \leq \omega \leq n-1$.

4.5.2 $\exists \rho_i=1$, and ρ_1 and every ρ_i are external, where $2 \leq i, j \leq k$ and $i \neq j$

For $k=3$, determining the length of each path is listed below. Each desired symbol in a correcting sequence takes 2 and a misplaced symbol takes 1. Analyzing lengths of vertex-disjoint paths in **i**–**iv** and **v** are as follows. Obviously, the length of **i** is 4; the length of **ii** is 5; the length of **iii** is 6; the length of **iv** is 6, and the length of **v** are 7. Two paths whose lengths are 6 should be further discussed. The $l(P(\rho, \rho^j, \varepsilon))$ with $\varepsilon_1^1 = \rho_g$ is at least 6, because $l(P^*(\rho^j, \varepsilon))$ is 5 by Lemma 4. The other path of length 6 is considering below. If $P^*(\rho, \rho^1, \varepsilon)$ with $\rho_1^1 = g$ and $P^*(\rho, \rho^g, \varepsilon)$ with $\varepsilon_1^1 = \rho_1$ are all shortest then both of them should contain ε^j . For retaining vertex-disjoint, one of them cannot contain ε^j and its length is at least 6. Paths in **v** with length 7 are considered below. Although the $l(P^*(\rho, \rho^1, \varepsilon))$ is 5 and $l(P^*(\rho, \rho^1, \varepsilon))$ with $\rho_1^1 = g$ is 4, these paths cannot avoid to contain ε^j . Hence, only one path can enclose ε^j , and thus the lengths of **v** are at least 7. Consequently, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,3}) = 5$ if $1 \leq \omega \leq 2$, $d_\omega(\rho, \varepsilon) = d(S_{n,3})+1 = 6$ if $3 \leq \omega \leq 4$, and $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,3})+2 = 7$ if $5 \leq \omega \leq n-1$.

For $k \geq 4$, calculating the length of each path is listed below. Recall that symbol $\rho_1^1 = x$.

i $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ for $2 \leq x \leq k-1$ with $\varepsilon_1^1 \neq \rho_1$ is $2k-2$

because $\rho^1 \rightarrow \rho^{1,x}$ takes 1; fixing $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.

- ii** $l(P(\rho, \rho^1, \varepsilon^{j+1}, \varepsilon))$ for $x=k$ is $2k-1$ because $l(P(\rho, \rho^{1,k,1,j \oplus 1}))$ takes 5, and fixing $k-3$ desired symbols takes $2k-6$.
- iii** $l(P(\rho, \rho^j, \varepsilon^1, \varepsilon))$ is $2k$ because $\rho \rightarrow \rho^j$ takes 1; fixing $k-1$ desired symbols takes $2k-2$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.
- iv** $l(P(\rho, \rho^j, \varepsilon^j, \varepsilon))$, where $i \neq j \oplus 1$, is $2k$ because $P(\rho, \rho^{i,1,j,1,i})$ takes 5, and fixing $k-3$ desired symbols and a misplaced symbol takes $2k-6+1$.
- v** $l(P(\rho, \rho^{j \oplus 1}, \varepsilon^j, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^{j \oplus 1}$ takes 1, and fixing $k-1$ desired symbols takes $2k-2$.
- vi** $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ is $2k+1$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j}$ takes 2; fixing $k-1$ desired symbols takes $2k-2$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.

There are k paths in **i**, **ii**, **v** whose lengths are $d(S_{n,k})$, $k-2$ paths in **iii** and **iv** whose lengths are $d(S_{n,k})+1$, and $n-2k+1$ paths in **vi** whose lengths are $d(S_{n,k})+2$. There are at most k paths of length no more than $d(S_{n,k})$ and one of them is a $P(\rho, \rho^j, \varepsilon)$. That the lengths of two or more $P(\rho, \rho^j, \varepsilon)$ s are no more than $d(S_{n,k})$ is impossible, because they will contain a same vertex which is a neighbors of ε . $P(\rho, \rho^1, \varepsilon)$ s whose lengths are at least $d(S_{n,k})+1$ because $l(P^*(\rho^1, \varepsilon))$ is $d(S_{n,k})$: fixing desired symbols takes $2k-2$, and fixing symbol 1 takes at last 1. If $\omega > k$ then exist a $P(\rho, \rho^j, \varepsilon)$ or $P(\rho, \rho^1, \varepsilon)$ and its length is thus more than $d(S_{n,k})$. If $\omega > 2k-2$ then at least one $P(\rho, \varepsilon^1, \varepsilon)$ is contained. Duo to a $P^*(\rho, \varepsilon^1, \varepsilon)$ has to contain two ε^1 s, $P(\rho, \varepsilon^1, \varepsilon)$ s with single ε^1 has a length at least $d(S_{n,k})+2$. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq k$, $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $k+1 \leq \omega \leq 2k-2$, and $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})+2$ if $2k-1 \leq \omega \leq n-1$.

4.5.3 Symbols of ρ exist one misplaced and else are external

Sections 3.5.3.1 and 3.5.3.2 present determination the lengths of $C_\omega(\rho, \varepsilon)$ s for ρ_1 is misplaced and not misplaced, respectively. Obviously, they also have a bottleneck when $k=3$, so $k=3$ is particular and $k \geq 4$ is general in each subsection.

4.5.3.1 Symbol ρ_1 is misplaced and every ρ_i are external, where $2 \leq i \leq k$

For $k=3$, computing the length of each path is stated as follows. Restated, each desired symbol in a correcting sequence takes 2 and a misplaced symbol takes 1. $l(P(\rho, \rho^1, \varepsilon))$ with $\rho_1^1 = 1$ is 6; $l(P(\rho, \rho^1, \varepsilon))$ with $\rho_1^1 = g$ is 5, $l(P(\rho, \rho^g, \varepsilon))$ is 6; $l(P(\rho, \rho^j, \varepsilon))$ is 4, and $l(P(\rho, \rho^1, \varepsilon))$ is 7. There are two paths whose lengths are 6. $l(P(\rho, \rho^1, \varepsilon))$ with $\rho_1^1 = 1$ is 6 because $l(P^*(\rho^1, \varepsilon))$ is 5 by Lemma 4. The other path with a length of 6 is considered below. Since $l(P^*(\rho, \rho^1, \varepsilon))$ with $\rho_1^1 = g$ and $l(P^*(\rho, \rho^g, \varepsilon))$ with $\varepsilon_1^1 = \rho_1$ should contain ε^1 with $\varepsilon_1^1 = \rho_1$, one path of them has to exclude ε^1 with $\varepsilon_1^1 = \rho_1$ and its length is at least 6. If

$\omega \geq 5$, a $P(\rho, \rho^1, \varepsilon)$ should be contained in a $C_\omega^*(\rho, \varepsilon)$, and $l(P(\rho, \rho^1, \varepsilon))$ is 7 by Lemma 5. $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,3}) = 5$ if $1 \leq \omega \leq 2$, $d_\omega(\rho, \varepsilon) = d(S_{n,3}) + 1 = 6$ if $3 \leq \omega \leq 4$, and $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,3}) + 2 = 7$ if $5 \leq \omega \leq n-1$.

For $k \geq 4$, computing the length of each path is presented as follows.

- i $l(P(\rho, \rho^1, \varepsilon^i, \varepsilon))$ for $x=1$ is $2k$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j}$ takes 2, and fixing $k-1$ desired symbols takes $2k-2$.
- ii $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ for $x=j \neq 1$ and $\varepsilon_1^1 = \rho_j$ is $2k-1$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j \oplus 1}$ takes 2; fixing $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.
- iii $l(P(\rho, \rho^1, \varepsilon^{x \oplus 1}, \varepsilon))$ for $2 \leq x \leq k$ and $x \neq j \oplus 1$ is $2k$ because $P(\rho, \rho^{1,x,1,x \oplus 1})$ takes 4, and fixing $k-2$ desired symbols takes $2k-4$.
- iv $l(P(\rho, \rho^j, \varepsilon^{j \oplus 1}, \varepsilon))$ is $2k-1$ because $P(\rho, \rho^{j,1,j \oplus 1})$ takes 3, and fixing $k-2$ desired symbols takes $2k-4$.
- v $l(P(\rho, \rho^i, \varepsilon^1, \varepsilon))$ for $i \neq j$ and $\varepsilon_1^1 \neq \rho_j$ is $2k-1$ because $\rho \rightarrow \rho^i$ takes 1; fixing $k-2$ desired symbols and a misplaced symbol takes $2k-4+1$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.
- vi $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ is $2k+1$ that can refer to Section 4.1.

There are k paths in ii, iv, and v whose lengths are $d(S_{n,k})$; $k-2$ paths in i and iii whose lengths are $d(S_{n,k})+1$, and $n-2k+1$ paths in vi whose lengths are $d(S_{n,k})+2$. There are at most k paths of length no more than $d(S_{n,k})$ and one of them is a $P(\rho, \varepsilon^i, \varepsilon)$. That the lengths of two or more $P(\rho, \varepsilon^i, \varepsilon)$ s are no more than $d(S_{n,k})$ is impossible, because they will contain a same vertex which is a neighbors of ρ . Hence, if $\omega > k$, then there exist two $P(\rho, \varepsilon^i, \varepsilon)$ s. The lengths of $P(\rho, \rho^1, \varepsilon)$ s are $d(S_{n,k})+2$ by Lemma 5. If $\omega > 2k-2$ then a $P(\rho, \rho^1, \varepsilon)$ should be contained in a $C_\omega^*(\rho, \varepsilon)$. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq k$, $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $k+1 \leq \omega \leq 2k-2$, and $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})+2$ if $2k-1 \leq \omega \leq n-1$.

4.5.3.2 $\exists \rho_j$ is misplaced, ρ_1 and every ρ_i are external for $2 \leq i, j \leq k$ and $i \neq j$

For $k=3$, analyzing the length of each path is demonstrated below. $l(P(\rho, \rho^1, \varepsilon))$ with $\rho_1^1=1$ is 5, and $l(P(\rho, \rho^1, \varepsilon))$ with $\rho_1^1=j$ is 4; $l(P(\rho, \rho^j, \varepsilon))$ is 5, and $l(P(\rho, \rho^j, \varepsilon))$ and $l(P(\rho, \rho^1, \varepsilon))$ are 6. Paths with a length of 6 are considered below. $P^*(\rho, \rho^j, \varepsilon)$ and $P^*(\rho, \rho^i, \varepsilon)$ should contain ε^1 which $\varepsilon_1^1 = \rho_1$, one path of them has to exclude ε^1 with $\varepsilon_1^1 = \rho_1$ and its length is at least 6. Referring to Section 4.2, $l(P(\rho, \rho^1, \varepsilon))$ is at least 6. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,3}) = 5$ if $1 \leq \omega \leq 3$ and $d_\omega(\rho, \varepsilon) = d(S_{n,3})+1 = 6$ if $4 \leq \omega \leq n-1$.

For $k \geq 4$, computing the length of each path is shown below. Referring to Section 4.2, $n-2k+1$ $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ are $d(S_{n,k})+1$. $l(P(\rho, \rho^1, \varepsilon^i, \varepsilon))$ with $x=1$ is $2k-1$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j}$ takes 2; fixing misplaced symbol ρ_j and $k-2$ desired symbols takes $1+2k-4$.

$l(P(\rho, \rho^1, \varepsilon^i, \varepsilon))$ with $x=j$ is $2k-1$ because $P(\rho, \rho^{1,x,y,1,s})$ takes 5 and $k-3$ desired symbols takes $2k-6$. $l(P(\rho, \rho^1, \varepsilon^i, \varepsilon))$, where $2 \leq x \leq k$ and $x \neq j$, is $2k-1$ because $P(\rho, \rho^{1,x,1,j,y,1,s})$ takes 7 and $k-4$ desired symbols takes $2k-8$. $l(P(\rho, \rho^j, \varepsilon^1, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^j \rightarrow \rho^{j,y}$ takes 2; $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. $l(P(\rho, \rho^{j \oplus 1}, \varepsilon^1, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^{j \oplus 1}$ takes 1; fixing misplaced symbol ρ_j and $k-2$ desired symbols takes $1+2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. $l(P(\rho, \rho^{j \oplus 1}, \varepsilon^1, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^{j \oplus 1}$ takes 1; fixing misplaced symbol ρ_j and $k-2$ desired symbols takes $1+2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq 2k-2$, and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $2k-3 \leq \omega \leq n-1$.

4.6 Case 2.3: $\rho_1 \neq 1$, and ρ has $k-2$ external symbols

According to Section 3.6, two cases are necessary to be considered as follows.

4.6.1 $\rho_1 = g$ is misplaced; $\exists \rho_j = 1$, and every ρ_i is external, where $2 \leq i, j, g \leq k$ and $i \neq j, g$

Recall that h represents the smallest symbol of i and l indicates an outer external symbol.

- i $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ with $\rho_1^1=x$, for $2 \leq x \leq k$, $x \notin \{\rho_1, j \oplus 1\}$ is $2k-1$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,x}$ takes 2; $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ with $\rho_1^1=j \oplus 1$ in Cases a, b and c are $2k-1$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j \oplus 1}$ takes 2; $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1. Thus, $k-2$ $P(\rho, \rho^1, \varepsilon^1, \varepsilon)$ s have a length of $d(S_{n,k})$.
- ii $l(P(\rho, \rho^g, \varepsilon^j, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^g$ takes 1 and fixing $k-1$ desired symbols takes $2k-2$.
- iii $l(P(\rho, \rho^j, \varepsilon^g, \varepsilon))$ is $2k-1$ because $\rho \rightarrow \rho^j \rightarrow \rho^{j,g}$ takes 2; fixing $k-2$ desired symbols and misplaced symbol g takes $2k-4+1$.
- iv $l(P(\rho, \rho^j, \varepsilon^h, \varepsilon))$ is $2k-2$ because $\rho \rightarrow \rho^j \rightarrow \rho^{j,h}$ takes 2 and fixing $k-2$ desired symbols takes $2k-4$.
- v $l(P(\rho, \rho^h, \varepsilon^j, \varepsilon))$ is $2k-2$ because $\rho \rightarrow \rho^h$ takes 1; fixing $k-2$ desired symbols and misplaced symbol j takes $2k-4+1$.
- vi $l(P(\rho, \rho^i, \varepsilon^j, \varepsilon))$ is $2k-1$ because $P(\rho, \rho^{i,1,j,i,g})$ takes 5 and fixing $k-3$ desired symbols takes $2k-6$. Notably, if $1 \leq \omega \leq 2k-3$ and $g=j$ then $i \neq h$ and only $k-3$ $P(\rho^i, \varepsilon^j)$ s can still be built. Thus, $k-3$ $P(\rho, \rho^i, \varepsilon^j, \varepsilon)$ s have a length of $d(S_{n,k})$. Similarly, for $2k-2 \leq \omega \leq n-1$ and $g=j$, the $k-3$ $P(\rho, \rho^i, \varepsilon^j, \varepsilon)$ s have a length of $d(S_{n,k})$.
- vii $l(P(\rho, \rho^g, \varepsilon^1, \varepsilon))$ with $\varepsilon_1^1=l$ is $2k$ because $P(\rho, \rho^{g,1,g-1})$ takes 3; fixing $k-2$ desired symbols takes $2k-4$, and $\varepsilon^1 \rightarrow \varepsilon$ takes 1.
- viii $l(P(\rho, \rho^1, \varepsilon^j, \varepsilon))$ with $\rho_1^1=l$ is $2k$ because $\rho \rightarrow \rho^1 \rightarrow \rho^{1,j-1}$ takes 2 and fixing $k-1$ desired symbols takes $2k-2$.
- ix Referring to Section 4.5.2, $l(P(\rho, \rho^1, \varepsilon^1, \varepsilon))$ is

$2k+1$. The lengths of these $n-2k+1$ $P(\rho, \rho^i, \varepsilon^i)$ s are $d(S_{n,k})+2$.

For $1 \leq \omega \leq 2k-3$, lengths of paths in **i**, **ii**, **iii**, **iv**, **v** and **vi** are no more than $d(S_{n,k})$. For $2k-2 \leq \omega \leq n-1$, a $P(\rho, \rho^i, \varepsilon)$ should be contained in a $C_\omega^*(\rho, \varepsilon)$. Referring to Section 4.5.2, only one $P(\rho, \rho^i, \varepsilon)$ containing ε^i has a length of $d(S_{n,k})+1$, and the other $P(\rho, \rho^i, \varepsilon)$ s excluding ε^i has a length of $d(S_{n,k})+2$. Consequently, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq 2k-3$; $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $\omega = 2k-2$, and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+2$ if $2k-1 \leq \omega \leq n-1$.

4.6.2 ρ_1 is external; $\exists \rho \neq 1$ and $\exists \rho_g$ is misplaced, $j \neq g$, and every ρ_i is external, where $2 \leq i, j, g \leq k$ and $i \notin \{j, g\}$

Since the numbers of ρ^1 and ρ^i are $k-2$ and $k-1$, respectively, $l(P(\rho, \rho^1, \varepsilon))$ and $l(P(\rho, \rho^i, \varepsilon))$ are no more than $d(S_{n,k})$ and can be referred to groups **i-iv** in Section 4.6.1. Moreover, $l(P(\rho, \rho^i, \varepsilon))$ is $d(S_{n,k})+1$ that can be proved as that in Case 1.2. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,k})$ if $1 \leq \omega \leq 2k-3$ and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+1$ if $2k-2 \leq \omega \leq n-1$.

5 ω -Wide Diameters

This section first discusses the special case for $k=2$ and then provides the main theorem of this work. All probable $\rho = \rho_1 \rho_2$ are $1e_2, 2e_2, 21, e_1e_2, e_11, \text{ and } e_12$. Let e' denote an outer external symbol. A $C_\omega^*(\rho, \varepsilon)$ can be built for each probable ρ and $l(C_\omega^*(\rho, \varepsilon))$ can also be determined by exhaustive search. When $\rho=21$, $C_\omega^*(\rho, \varepsilon)$ is the longest one among all $C_\omega^*(\rho, \varepsilon)$ s. Thus, $l(C_\omega^*(\rho, \varepsilon))$ for $\rho=21$ equals to $d_\omega(S_{n,2})$ for $2 \leq \omega \leq n-1$. The $C_\omega^*(\rho, \varepsilon)$ for $\rho=21$ is constructed as $21(\rho) \rightarrow 12(\varepsilon), 21(\rho) \rightarrow e'1(\rho^1) \rightarrow 1e'(\rho^{1,2}) \rightarrow 2e'(\rho^{1,2,1}) \rightarrow e'2(\rho^{1,2,1,2}) \rightarrow 12(\varepsilon)$ in which one path has a length of 1 and $n-2$ paths have a length of 5 ($d(S_{n,k})+2$). Apparently, the constructed paths are vertex-disjoint. The $l(P(\rho, \rho^1, \varepsilon))$ is 5, because $e'1(\rho^1)$ has to go through $1e'(\rho^{1,2})$ for vertex-disjoint, and $1e'(\rho^{1,2})$ to ε takes $d(S_{n,k})$ by Lemma 4. Therefore, $d_\omega(\rho, \varepsilon) = l(C_\omega^*(\rho, \varepsilon)) = d(S_{n,2}) = 3$ if $\omega = 1$, and $d_\omega(\rho, \varepsilon) = d(S_{n,k})+2 = 5$ if $2 \leq \omega \leq n-1$.

Based on the results described in Sections 4 and 5, Theorem 6 can thus be obtained as follows.

Theorem 6. $d_\omega(S_{n,2}) = d(S_{n,2})+2$ for $2 \leq \omega \leq n-1$, $d_\omega(S_{n,k}) = d(S_{n,k})$ for $2 \leq \omega \leq k-1$ and $3 \leq k \leq \lfloor n/2 \rfloor$, $d_\omega(S_{n,k}) = d(S_{n,k})+1$ for $k-1 < \omega \leq 2k-2$ and $3 \leq k \leq \lfloor n/2 \rfloor$, $d_\omega(S_{n,k}) = d(S_{n,k})+2$ for $2k-2 < \omega \leq n-1$ and $3 \leq k \leq \lfloor n/2 \rfloor$.

6 Conclusion

Constructing vertex-disjoint paths and computing ω -wide diameters of a graph are very important issues for evaluating reliability, fault-tolerant ability,

maximum parallelism and minimum transmission delay of the graph. This work provides routing algorithms to build a $C_\omega^*(\rho, \varepsilon)$ for each selected vertex ρ in an $S_{n,k}$ based on the $C_{n-1}(\rho, \varepsilon)$ constructed by Lin and Duh. After computing the length of each constructed $C_\omega^*(\rho, \varepsilon)$, $d_\omega(S_{n,k})$ for $2 \leq k \leq \lfloor n/2 \rfloor$ can thus be determined. These measurement results demonstrate that $d_\omega(S_{n,2}) = d(S_{n,2})+2$ for $2 \leq \omega \leq n-1$, $d_\omega(S_{n,k}) = d(S_{n,k})$ for $2 \leq \omega \leq k-1$ and $3 \leq k \leq \lfloor n/2 \rfloor$, $d_\omega(S_{n,k}) = d(S_{n,k})+1$ for $k-1 < \omega \leq 2k-2$ and $3 \leq k \leq \lfloor n/2 \rfloor$, $d_\omega(S_{n,k}) = d(S_{n,k})+2$ for $2k-2 < \omega \leq n-1$ and $3 \leq k \leq \lfloor n/2 \rfloor$. Future work will propose algorithms to construct $C_\omega^*(\rho, \varepsilon)$ in an $S_{n,k}$ and compute $d_\omega(S_{n,k})$ when $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$.

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